INCH-POUND

MIL-M-28787/428
26 November 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MIL-STD-1553 SUBSYSTEM ADAPTER CARD MODULE

KEY CODE GWA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic MIL-STD-1553 Subsystem Adapter Card (SAC) module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 3.496 failures per million operating hours for class I or 10.842 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be GWA.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	O	----------Module shall pass all remaining tests----------						
D1	Ibd	V	S	0.0	0.5	0.0	0.5	0.0	0.5	µA
D2	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	µA
D3-D5	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	µA
D6, D7	Ibd	V	S	0.0	1.2	0.0	1.2	0.0	1.2	µA
D8	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	µA
D9	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	µA
D10	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	µA
D11	Ibd	V	S	0.0	3.0	0.0	3.0	0.0	3.0	µA
D12	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	µA
D13	Ibd	V	S	-0.024	2.02	-0.024	2.02	-0.024	2.02	µA
D14	Ibd	V	S	0.0	2.1	0.0	2.1	0.0	2.1	µA
D15	Ibd	V	S	0.0	4.0	0.0	4.0	0.0	4.0	µA
D16	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	µA
D17	Ibd	V	S	0.0	2.2	0.0	2.2	0.0	2.2	µA
D18	Ibd	V	S	0.0	2.1	0.0	2.1	0.0	2.1	µA
D19, D20	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	µA
D21	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	µA
D22-D25	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	µA
D26	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	µA
D27	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	µA
D28	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	µA
D29	Ibd	V	S	-0.024	3.02	-0.024	3.02	-0.024	3.02	µA
D30	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	µA
D31	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	µA
D32-D36	Ibd	V	S	0.0	0.005	0.0	0.005	0.0	0.005	µA
D37	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	µA
D38-D42	Ibd	V	S	0.0	0.005	0.0	0.005	0.0	0.005	µA
D43-D45	Ibd	V	S	0.0	0.01	0.0	0.01	0.0	0.01	µA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
D46,D47	Ibd	V	S	0.0	0.005	0.0	0.005	0.0	0.005	mA
D48	Ibd	V	S	0.0	2.005	0.0	2.005	0.0	2.005	mA
D49,D50	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
E1	Ios	VI	S	-250.0	-30.0	-250.0	-30.0	-250.0	-30.0	mA
E2-E6	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E7,E8	Ios	VI	S	-250.0	-30.0	-250.0	-30.0	-250.0	-30.0	mA
E9	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E10-E12	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E13	Ios	VI	S	-220.0	-60.0	-220.0	-60.0	-220.0	-60.0	mA
E14-E47	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
F1	Ith	VII	X	-3.0	0.1	-3.0	0.1	-3.0	0.1	mA
F2	Ith	VII	X	-0.85	0.045	-0.85	0.045	-0.85	0.045	mA
F3-F5	Ith	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
F6,F7	Ith	VII	X	-0.855	0.05	-0.855	0.05	-0.855	0.05	mA
F8	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F9	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
F10	Ith	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
F11	Ith	VII	X	-0.75	0.075	-0.75	0.075	-0.75	0.075	mA
F12	Ith	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
F13	Ith	VII	X	-1.65	-0.562	-1.65	-0.562	-1.65	-0.562	mA
F14	Ith	VII	X	-0.7	0.07	-0.7	0.07	-0.7	0.07	mA
F15	Ith	VII	X	-1.0	0.1	-1.0	0.1	-1.0	0.1	mA
F16	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
F17	Ith	VII	X	-1.1	0.09	-1.1	0.09	-1.1	0.09	mA
F18	Ith	VII	X	-0.9	0.07	-0.9	0.07	-0.9	0.07	mA
F19,F20	Ith	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
F21	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
F22-F25	Ith	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
F26	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
F27	Ith	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
F28	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
F29	Ith	VII	X	-1.9	-0.537	-1.9	-0.537	-1.9	-0.537	mA

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F30	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	A
F31	Ith	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	A
F32-F38	Ith	VII	X	-0.02	0.02	-0.02	0.02	-0.02	0.02	A
F38-F43	Ith	VII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	A
F44	Ith	VII	X	-0.02	0.02	-0.02	0.02	-0.02	0.02	A
F45	Ith	VII	X	-0.23	0.05	-0.23	0.05	-0.23	0.05	A
F46,F47	Ith	VII	X	-0.03	0.03	-0.03	0.03	-0.03	0.03	A
F48	Ith	VII	X	-0.23	0.05	-0.23	0.05	-0.23	0.05	A
F49-F52	Ith	VII	X	-0.03	0.03	-0.03	0.03	-0.03	0.03	A
F53	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	A
F54-F58	Ith	VII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	A
F59-F61	Ith	VII	X	-0.01	0.01	-0.01	0.01	-0.01	0.01	A
F62,F63	Ith	VII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	A
F64	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	A
F65,F66	Ith	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	A
G1	Iil	VIII	X	-3.0	0.1	-3.0	0.1	-3.0	0.1	A
G2	Iil	VIII	X	-0.85	0.045	-0.85	0.045	-0.85	0.045	A
G3-G5	Iil	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	A
G6,G7	Iil	VIII	X	-0.855	0.05	-0.855	0.05	-0.855	0.05	A
G8	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	A
G9	Iil	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	A
G10	Iil	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	A
G11	Iil	VIII	X	-0.75	0.075	-0.75	0.075	-0.75	0.075	A
G12	Iil	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	A
G13	Iil	VIII	X	-1.65	-0.562	-1.65	-0.562	-1.65	-0.562	A
G14	Iil	VIII	X	-0.7	0.07	-0.7	0.07	-0.7	0.07	A
G15	Iil	VIII	X	-1.0	0.1	-1.0	0.1	-1.0	0.1	A
G16	Iil	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	A
G17	Iil	VIII	X	-1.1	0.09	-1.1	0.09	-1.1	0.09	A
G18	Iil	VIII	X	-0.9	0.07	-0.9	0.07	-0.9	0.07	A
G19,G20	Iil	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	A
G21	Iil	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	A
G22-G25	Iil	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	A

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G26	Iil	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
G27	Iil	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
G28	Iil	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
G29	Iil	VIII	X	-1.9	-0.537	-1.9	-0.537	-1.9	-0.537	mA
G30	Iil	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
G31	Iil	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
G32-G38	Iil	VIII	X	-0.02	0.02	-0.02	0.02	-0.02	0.02	mA
G39-G43	Iil	VIII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	mA
G44	Iil	VIII	X	-0.02	0.02	-0.02	0.02	-0.02	0.02	mA
G45	Iil	VIII	X	-0.23	0.05	-0.23	0.05	-0.23	0.05	mA
G46,G47	Iil	VIII	X	-0.03	0.03	-0.03	0.03	-0.03	0.03	mA
G48	Iil	VIII	X	-0.23	0.05	-0.23	0.05	-0.23	0.05	mA
G49-G52	Iil	VIII	X	-0.03	0.03	-0.03	0.03	-0.03	0.03	mA
G53	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G54-G58	Iil	VIII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	mA
G59-G61	Iil	VIII	X	-0.01	0.01	-0.01	0.01	-0.01	0.01	mA
G62,G63	Iil	VIII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	mA
G64	Iil	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
G65,G66	Iil	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
H1-H47	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J48	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K8	Idzh	XI	X	-0.01	0.01	-0.01	0.01	-0.01	0.01	mA
K9-K16	Idzl	XI	X	-0.01	0.01	-0.01	0.01	-0.01	0.01	mA
L1,L2	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L3-L5	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L6-L8	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L9-L29	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L30-L45	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L46-L48	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V

- See footnotes at end of table.

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				Min	Max	Min	Max	Min	Max	
M1	PSFC	XIII	S	11.0	20.5	11.0	20.5	11.0	20.5	uF
M2	PSFC	XIII	S	7.5	13.9	7.5	13.9	7.5	13.9	uF
N	Ipo	XIV	X	900	2200	800	2200	900	2200	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R1	C1/D	XVIII	Q	8	125	8	125	8	125	pF
R2	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R3-R5	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R6,R7	C1/D	XVIII	Q	4	150	4	150	4	150	pF
R8	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R9	C1/D	XVIII	Q	6	150	6	150	6	150	pF
R10	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R11	C1/D	XVIII	Q	6	100	6	100	6	100	pF
R12	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R13	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R14	C1/D	XVIII	Q	6	100	6	100	6	100	pF
R15	C1/D	XVIII	Q	4	125	4	125	4	125	pF
R16	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R17	C1/D	XVIII	Q	8	125	8	125	8	125	pF
R18	C1/D	XVIII	Q	6	125	6	125	6	125	pF
R19,R20	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R21	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R22-R25	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R26	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R27	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R28	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R29	C1/D	XVIII	Q	6	75	6	75	6	75	pF
R30	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R31	C1/D	XVIII	Q	2	75	2	75	2	75	pF

- See footnotes at end of table.

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				Min	Max	Min	Max	Min	Max	
R32-R38	C1/a	XVIII	Q	6	100	6	100	6	100	pf
R38-R43	C1/a	XVIII	Q	2	75	2	75	2	75	pf
R44	C1/a	XVIII	Q	6	100	6	100	6	100	pf
R45	C1/a	XVIII	Q	2	175	2	175	2	175	pf
R46	C1/a	XVIII	Q	6	125	6	125	6	125	pf
R47	C1/a	XVIII	Q	6	100	6	100	6	100	pf
R48	C1/a	XVIII	Q	2	175	2	175	2	175	pf
R49-R52	C1/a	XVIII	Q	6	100	6	100	6	100	pf
R53-R58	C1/a	XVIII	Q	2	75	2	75	2	75	pf
R59-R61	C1/a	XVIII	Q	4	75	4	75	4	75	pf
R62-R64	C1/a	XVIII	Q	2	75	2	75	2	75	pf
R65	C1/a	XVIII	Q	2	50	2	50	2	50	pf
S1,S2	Vio latch up	XIX	Q	-1.2	2.4	-1.2	2.4	-1.2	2.4	V
T1,T2	PSTA'	XX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/428-1 for class I and M28787/428-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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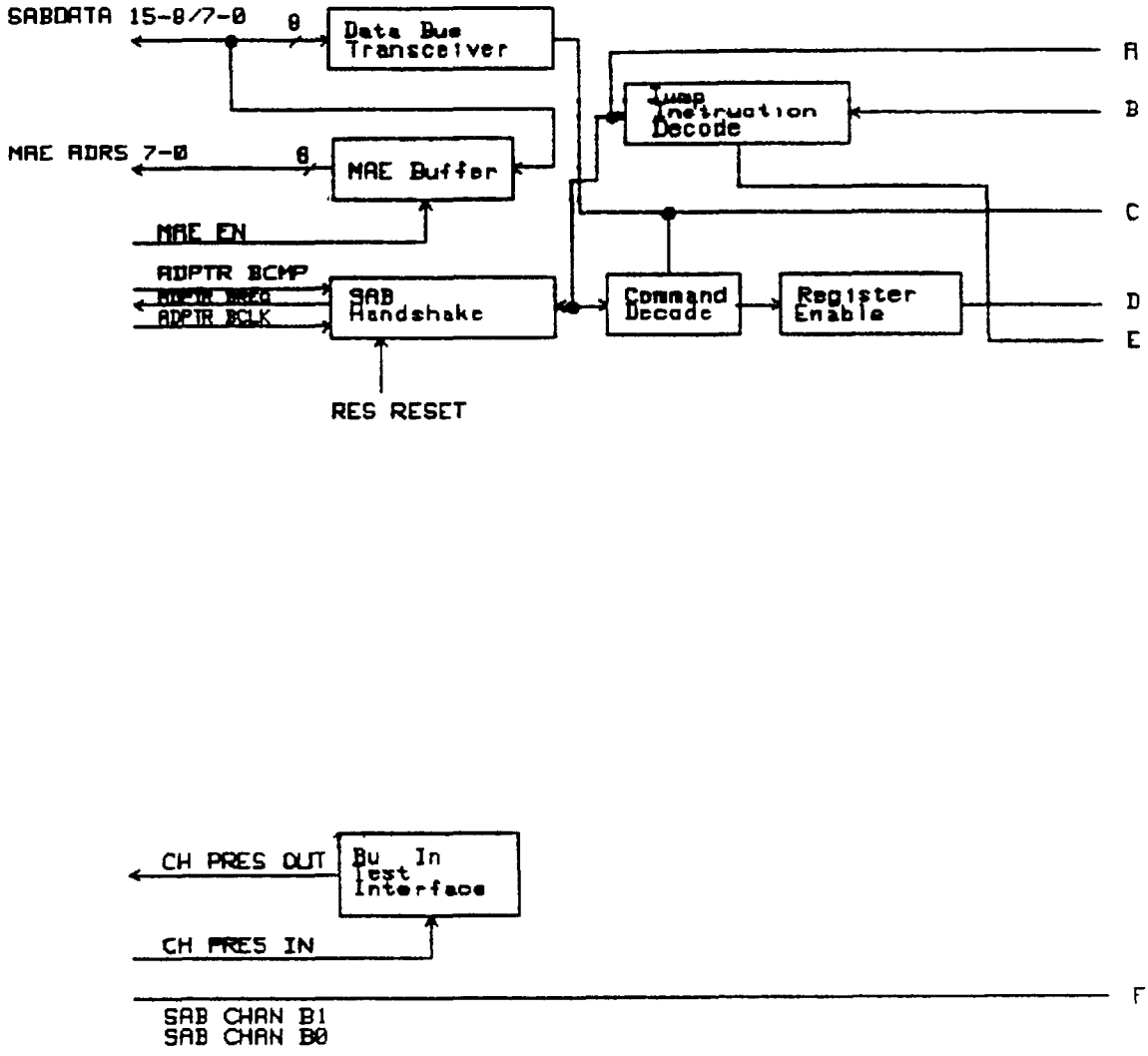


FIGURE 1. GWA functional block diagram.

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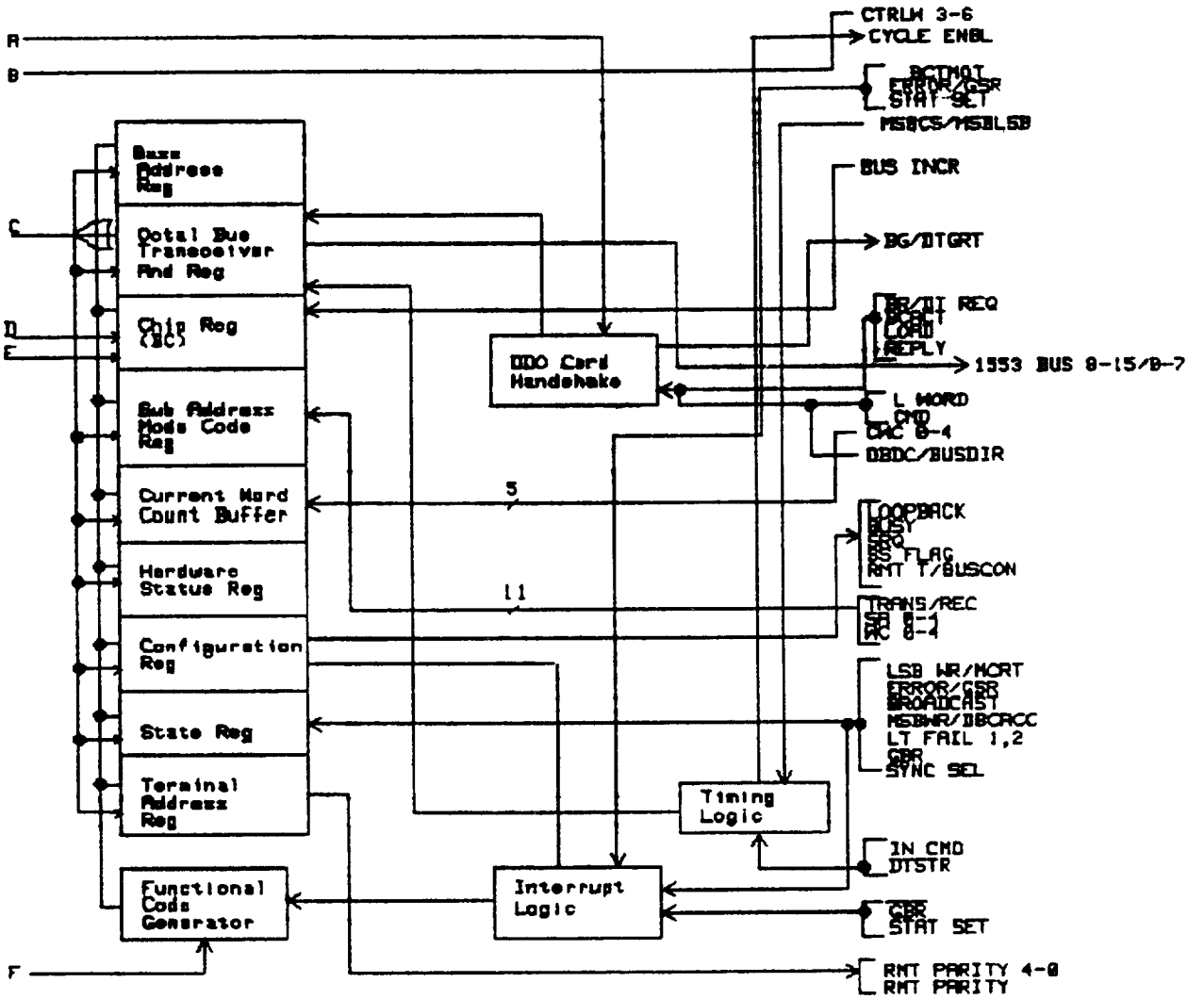


FIGURE 1. GWA functional block diagram - Continued.

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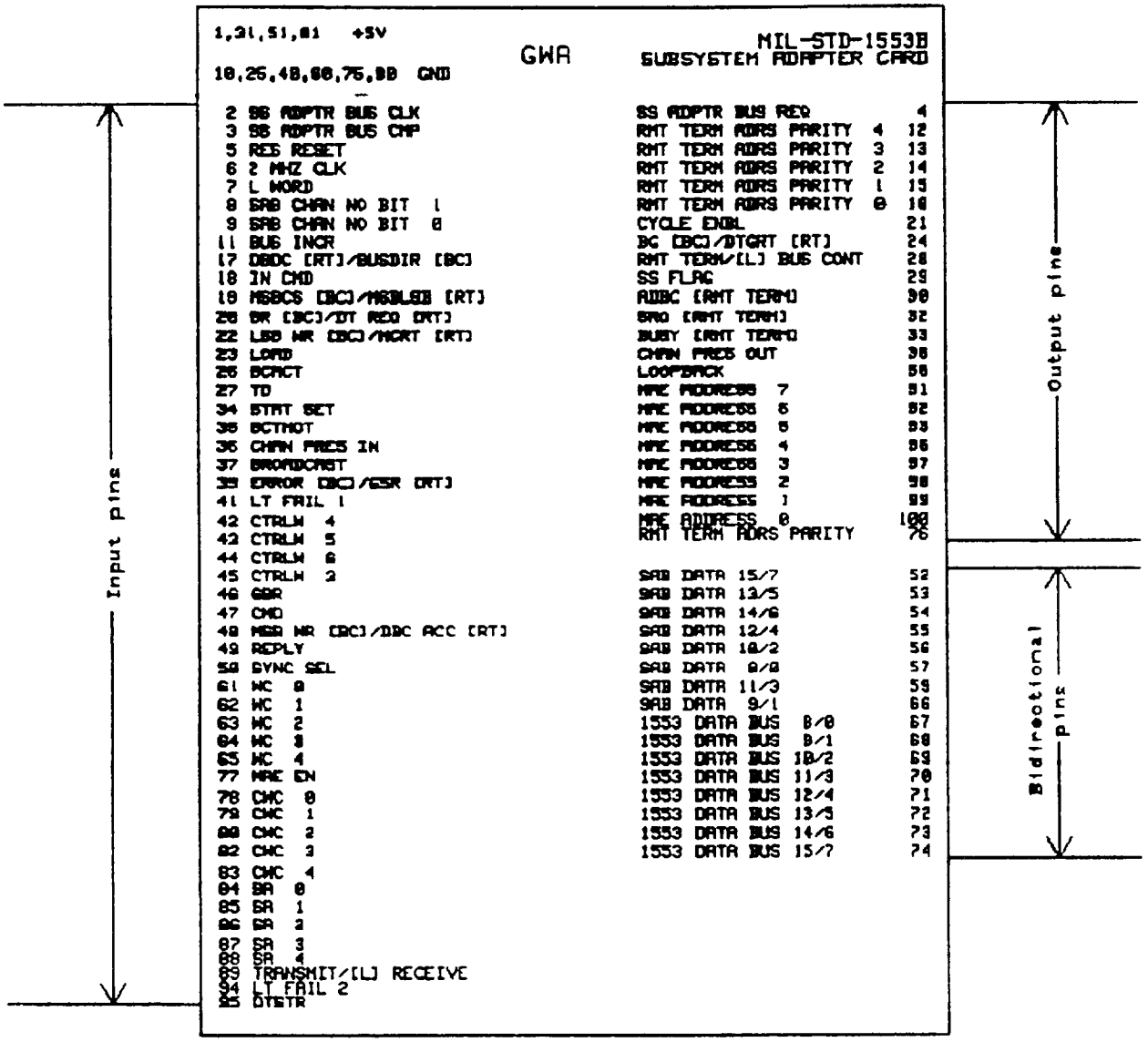


FIGURE 2. GWA pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, L, and S are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E, L, or S are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, L, and S where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

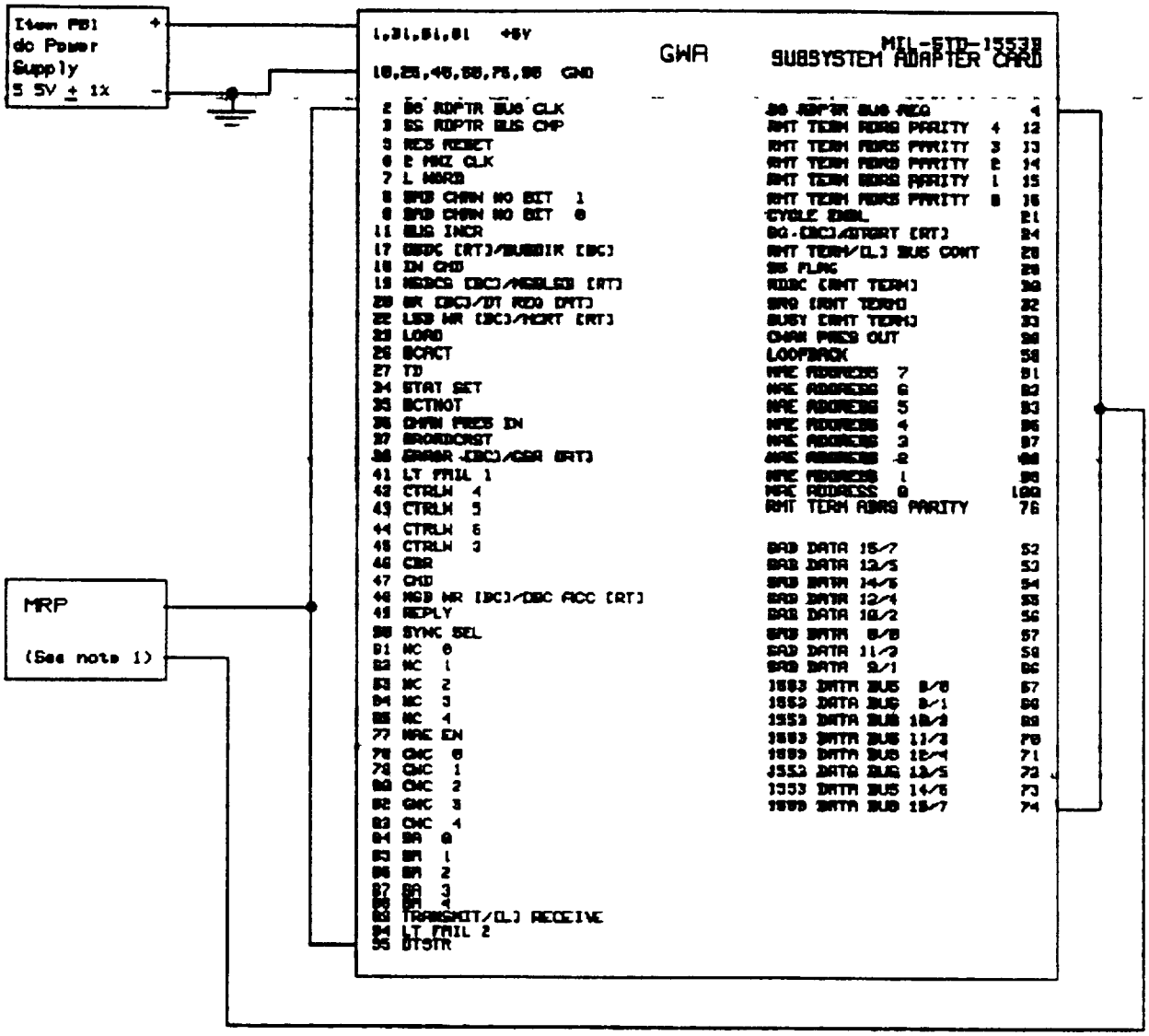
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. GWA operating temperature and component temperature test circuit.

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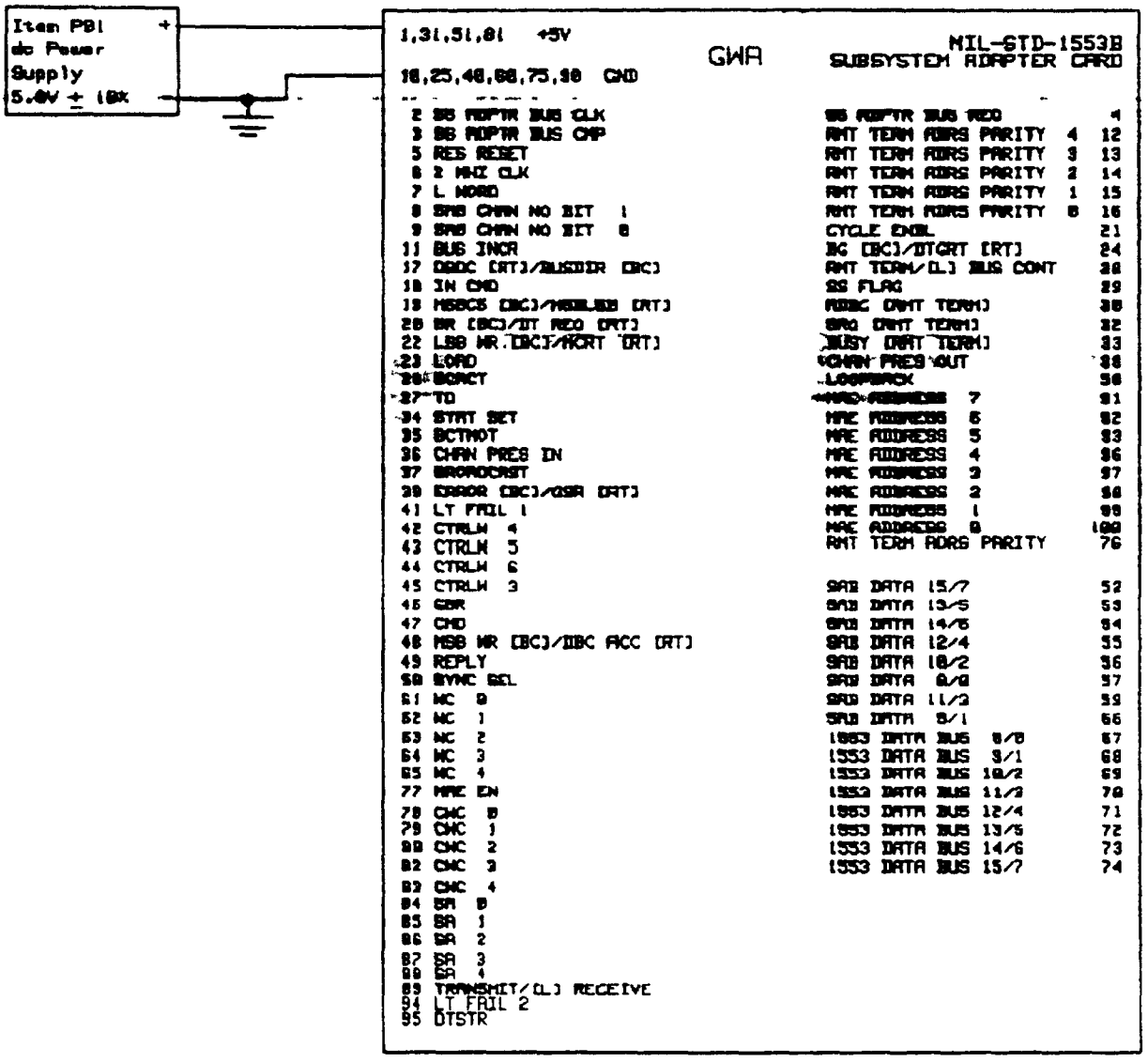


FIGURE 4. GWA vibration test circuit.

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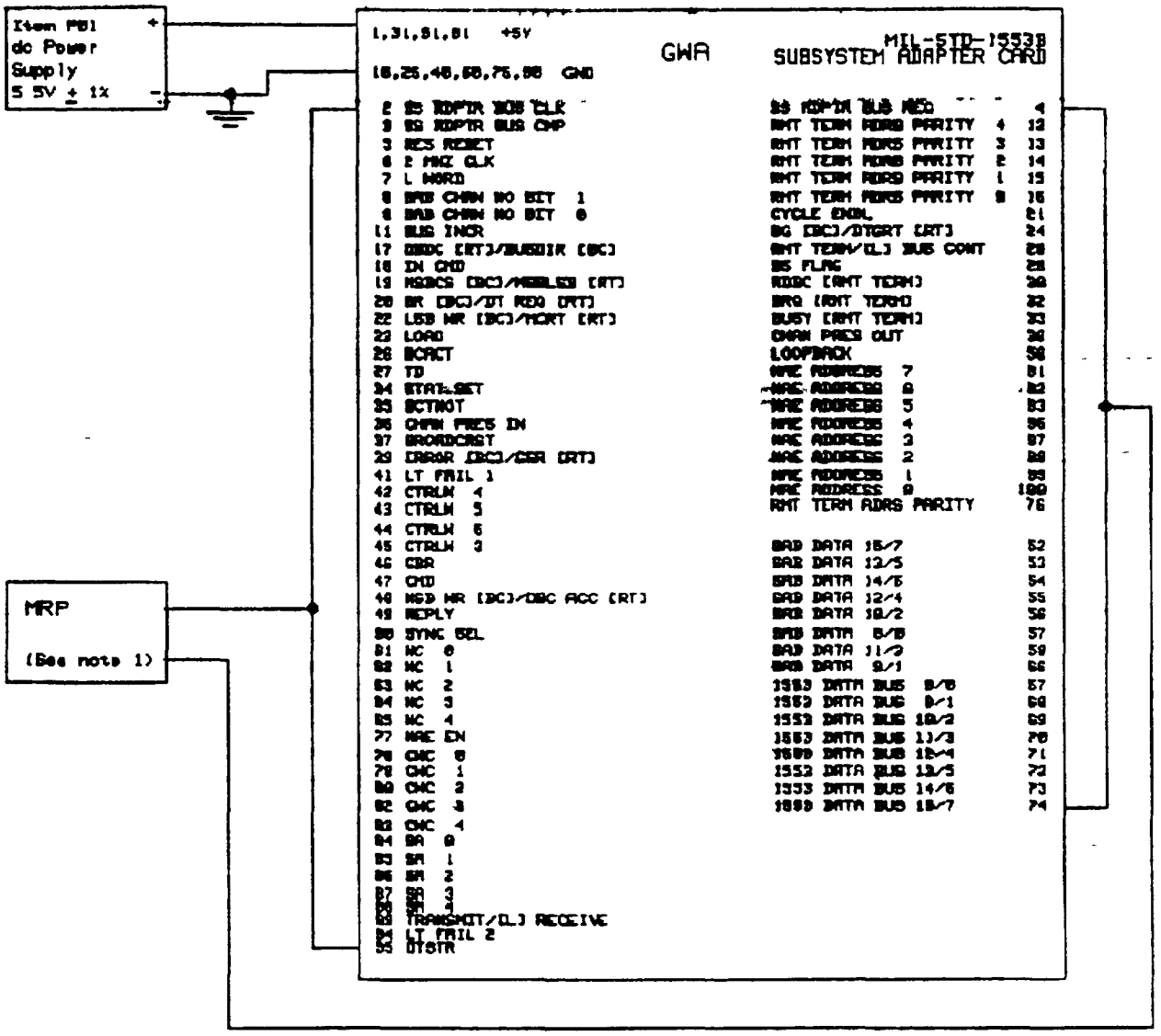
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. GWA life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 12.1 W (max); 8.075 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
ADPTR BCLK	-1.2	0.4	2.4	-3.0	0.1	0.5	125
ADPTR BCMP	-1.2	0.4	2.4	-0.85	0.045	1.1	75
RES RESET	-1.5	0.4	2.4	-0.25	0.025	1.0	75
2 MHZ CLK	-1.5	0.4	2.4	-0.25	0.025	1.0	75
L WORD	-1.5	0.4	2.4	-0.25	0.025	1.0	75
SAB CHAN B1	-1.2	0.4	2.4	-0.855	0.05	1.2	150
SAB CHAN B0	-1.2	0.4	2.4	-0.855	0.05	1.2	150
BUS INCR	-1.2	0.4	2.4	-0.6	0.02	0.1	75
DBDC/BUSDIR	-1.5	0.4	2.4	-0.5	0.05	2.0	150
IN CMD	-1.5	0.4	2.4	-0.25	0.025	1.0	50
MSBCS/MSBLSB	-1.5	0.4	2.4	-0.75	0.075	3.0	100
BR/DT REQ	-1.5	0.4	2.4	-0.25	0.025	1.0	75
LSB WR/MCRT	NA	0.4	2.4	-1.65	-0.562	2.02	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
LOAD	-1.5	0.4	2.4	-0.7	0.07	2.1	100
BCACT	-1.5	0.4	2.4	-1.0	0.1	4.0	125
TD	-1.5	0.4	2.4	-0.5	0.05	2.0	75
STAT SET	-1.5	0.4	2.4	-1.1	0.09	2.2	125
BCTMOT	-1.5	0.4	2.4	-0.9	0.07	2.1	125
CH PRES IN	-1.5	0.4	2.4	-0.25	0.025	1.0	50
BROADCAST	-1.5	0.4	2.4	-0.25	0.025	1.0	50
ERROR/GSR	-1.5	0.4	2.4	-0.5	0.05	2.0	100
LT FAIL 1	-1.5	0.4	2.4	-0.25	0.025	1.0	75
CTRLW 4	-1.5	0.4	2.4	-0.25	0.025	1.0	75
CTRLW 5	-1.5	0.4	2.4	-0.25	0.025	1.0	75
CTRLW 6	-1.5	0.4	2.4	-0.25	0.025	1.0	75
CTRLW 3	-1.5	0.4	2.4	-0.5	0.05	2.0	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
GBR	-1.5	0.4	2.4	-0.25	0.025	1.0	75
CMD	-1.5	0.4	2.4	-0.5	0.05	2.0	75
MSBWR/DBCACC	NA	0.4	2.4	-1.9	-0.537	3.02	75
REPLY	-1.5	0.4	2.4	-0.5	0.05	2.0	100
SYNC SEL	-1.5	0.4	2.4	-0.25	0.025	1.0	75
SABDATA 15/7	NA	0.4	2.4	-0.02	0.02	NA	100
SABDATA 13/5	NA	0.4	2.4	-0.02	0.02	NA	100
SABDATA 14/6	NA	0.4	2.4	-0.02	0.02	NA	100
SABDATA 12/4	NA	0.4	2.4	-0.02	0.02	NA	100
SABDATA 10/2	NA	0.4	2.4	-0.02	0.02	NA	100
SABDATA 8/0	NA	0.4	2.4	-0.02	0.02	NA	100
SABDATA 11/3	NA	0.4	2.4	-0.02	0.02	NA	100
WC 0	-1.2	0.4	2.4	-0.005	0.005	0.005	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
WC 1	-1.2	0.4	2.4	-0.005	0.005	0.005	75
WC 2	-1.2	0.4	2.4	-0.005	0.005	0.005	75
WC 3	-1.2	0.4	2.4	-0.005	0.005	0.005	75
WC 4	-1.2	0.4	2.4	-0.005	0.005	0.005	75
SABDATA 9/1	NA	0.4	2.4	-0.02	0.02	NA	100
1553BUS 8/0	NA	0.4	2.4	-0.23	0.05	NA	175
1553BUS 9/1	NA	0.4	2.4	-0.03	0.03	NA	125
1553BUS 10/2	NA	0.4	2.4	-0.03	0.03	NA	100
1553BUS 11/3	NA	0.4	2.4	-0.23	0.05	NA	175
1553BUS 12/4	NA	0.4	2.4	-0.03	0.03	NA	100
1553BUS 13/5	NA	0.4	2.4	-0.03	0.03	NA	100
1553BUS 14/6	NA	0.4	2.4	-0.03	0.03	NA	100
1553BUS 15/7	NA	0.4	2.4	-0.03	0.03	NA	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MAE EN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CWC 0	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CWC 1	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CWC 2	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CWC 3	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CWC 4	-1.2	0.4	2.4	-0.005	0.005	0.005	75
SA 0	-1.2	0.4	2.4	-0.01	0.01	0.01	75
SA 1	-1.2	0.4	2.4	-0.01	0.01	0.01	75
SA 2	-1.2	0.4	2.4	-0.01	0.01	0.01	75
SA 3	-1.2	0.4	2.4	-0.005	0.005	0.005	75
SA 4	-1.2	0.4	2.4	-0.005	0.005	0.005	75
TRANS/REC	-1.5	0.4	2.4	-0.5	0.05	2.005	75
LT FAIL 2	-1.5	0.4	2.4	-0.25	0.025	1.0	NA
DTSTR	-1.5	0.4	2.4	-0.25	0.025	1.0	50

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
ADPTR BREQ	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
RMT PARITY 4	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
RMT PARITY 3	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
RMT PARITY 2	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
RMT PARITY 1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
RMT PARITY 0	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
CYCLE ENBL	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
BG/DTGRT	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
RMT T/BUSCON	2.4	0.5	-0.16	1.6	NA	NA	-60.0	NA
SS FLAG	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ADBC	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
SRQ	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
BUSY	2.4	0.5	-0.16	1.6	NA	NA	-60.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
CH PRES OUT	2.4	0.5	-0.16	1.6	NA	NA	NA	NA
SABDATA 15/7	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
SABDATA 13/5	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
SABDATA 14/6	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
SABDATA 12/4	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
SABDATA 10/2	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
SABDATA 8/0	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
LOOPBACK	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
SABDATA 11/3	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
SABDATA 9/1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
1553BUS 8/0	2.4	0.5	-0.4	6.0	NA	NA	-60.0	175
1553BUS 9/1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125
1553BUS 10/2	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
1553BUS 11/3	2.4	0.5	-0.4	6.0	NA	NA	-60.0	175
1553BUS 12/4	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
1553BUS 13/5	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
1553BUS 14/6	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
1553BUS 15/7	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
RMT PARITY	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
MAE ADRS 7	2.4	0.5	-0.4	6.0	0.01	0.01	-60.0	NA
MAE ADRS 6	2.4	0.5	-0.4	6.0	0.01	0.01	-60.0	NA
MAE ADRS 5	2.4	0.5	-0.4	6.0	0.01	0.01	-60.0	NA
MAE ADRS 4	2.4	0.5	-0.4	6.0	0.01	0.01	-60.0	NA
MAE ADRS 3	2.4	0.5	-0.4	6.0	0.01	0.01	-60.0	NA
MAE ADRS 2	2.4	0.5	-0.4	6.0	0.01	0.01	-60.0	NA
MAE ADRS 1	2.4	0.5	-0.4	6.0	0.01	0.01	-60.0	NA
MAE ADRS 0	2.4	0.5	-0.4	6.0	0.01	0.01	-60.0	NA

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	th	ADPTR BCLK to SABDATA 15-8/7-0	100	100	100	-	-	-	nS	12
2	tsu	BR/DT REQ to ADPTR BCLK	91	91	91	-	-	-	nS	12
3	tpd	ADPTR BCLK to ADPTR BREQ	-	-	-	30	30	30	nS	12
4	tpd	ADPTR BCLK to BG/DTGRT	-	-	-	78	78	78	nS	12
5	tpd	ADPTR BCLK to 1553BUS 8-9/0-1	-	-	-	46	46	46	nS	12
5	tpd	ADPTR BCLK to 1553BUS 11-15/3-7	-	-	-	46	46	46	nS	12
6	tpd	LSB WR/MCRT to BUSY	-	-	-	78	78	78	nS	12
7	tpd	MAE EN to MAE ADRS 7-0	-	-	-	18	18	18	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	ADPTR BCLK to SRQ	-	-	-	57	57	57	nS	12
9	tpd	ADPTR BCLK to CYCLE ENBL	-	-	-	45	45	45	nS	12
10	tsu	SABDATA 15-8/7-0 to ADPTR BCLK	79	79	79	-	-	-	nS	12

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6.4 Module function. The Subsystem Adapter Card (SAC) accepts incoming data/mode codes from a set of interface Standard Electronic Modules (SEMs): Bus Controller (key code UKS), Dumb Remote Terminal Unit Protocol (key code UKT), and Remote Terminal Unit (RTU) Protocol (key code UKU) that function as a dual redundant bus controller or as remote terminal channels. The SAC generates the appropriate handshake signals to transfer data to the Common Bus Interface (CBI) module (key code GWB). The SAC stores pertinent information about the current status of the MIL-STD-1553B channel in a state register which is accessible to the Central Processor (CP). The module set is software configurable through the use of a configuration register. Interrupts are generated after various bus messages, errors or programmable conditions. The SAC posts interrupts by first making a request for the subsystem bus. When a grant has been received, the SAC transmits a specific code to the CBI that indicates the priority of the interrupt request. The cause of the interrupt is stored in the state register.

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6.4.1 Description of I/O pins.

Signal name	Pin number	I/O	Description
ADPTR BCLK	2	I	SS ADPTR BUS CLK
ADPTR BCMP	3	I	SS ADPTR BUS CMP
RES RESET	5	I	RES RESET
2 MHZ CLK	6	I	2 MHZ CLK
L WORD	7	I	L WORD
SAB CHAN B1	8	I	SAB CHAN NO BIT 1
SAB CHAN B0	9	I	SAB CHAN NO BIT 0
BUS INCR	11	I	BUS INCR
DBDC/BUSDIR	17	I	DBDC [RT]/BUSDIR [BC]
IN CMD	18	I	IN CMD
MSBCS/MSBLSB	19	I	MSBCS [BC]/MSBLSB [RT]
BR/DT REQ	20	I	BR [BC]/DT REQ [RT]
LSB WR/MCRT	22	I	LSB WR [BC]/MCRT [RT]
LOAD	23	I	LOAD
BCACT	26	I	BCACT
TD	27	I	TD
STAT SET	34	I	STAT SET
BCTMOT	35	I	BCTMOT
CH PRES IN	36	I	CHAN PRES IN
BROADCAST	37	I	BROADCAST

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
ERROR/GSR	39	I	ERROR [BC]/GSR [RT]
LT FAIL 1	41	I	LT FAIL 1
CTRLW 4	42	I	CTRLW 4
CTRLW 5	43	I	CTRLW 5
CTRLW 6	44	I	CTRLW 6
CTRLW 3	45	I	CTRLW 3
GBR	46	I	GBR
CMD	47	I	CMD
MSBWR/DBCACC	48	I	MSB WR [BC]/DBC ACC [RT]
REPLY	49	I	REPLY
SYNC SEL	50	I	SYNC SEL
SABDATA 15/7	52	I/O	SAB DATA 15/7
SABDATA 13/5	53	I/O	SAB DATA 13/5
SABDATA 14/6	54	I/O	SAB DATA 14/6
SABDATA 12/4	55	I/O	SAB DATA 12/4
SABDATA 10/2	56	I/O	SAB DATA 10/2
SABDATA 8/0	57	I/O	SAB DATA 8/0
SABDATA 11/3	59	I/O	SAB DATA 11/3
WC 0	61	I	WC 0
WC 1	62	I	WC 1

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
WC 2	63	I	WC 2
WC 3	64	I	WC 3
WC 4	65	I	WC 4
SABDATA 9/1	66	I/O	SAB DATA 9/1
1553BUS 8/0	67	I/O	1553 DATA BUS 8/0
1553BUS 9/1	68	I/O	1553 DATA BUS 9/1
1553BUS 10/2	69	I/O	1553 DATA BUS 10/2
1553BUS 11/3	70	I/O	1553 DATA BUS 11/3
1553BUS 12/4	71	I/O	1553 DATA BUS 12/4
1553BUS 13/5	72	I/O	1553 DATA BUS 13/5
1553BUS 14/6	73	I/O	1553 DATA BUS 14/6
1553BUS 15/7	74	I/O	1553 DATA BUS 15/7
MAE EN	77	I	MAE EN
CWC 0	78	I	CWC 0
CWC 1	79	I	CWC 1
CWC 2	80	I	CWC 2
CWC 3	82	I	CWC 3
CWC 4	83	I	CWC 4
SA 0	84	I	SA 0
SA 1	85	I	SA 1

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
SA 2	86	I	SA 2
SA 3	87	I	SA 3
SA 4	88	I	SA 4
TRANS/REC	89	I	TRANSMIT/[L] RECEIVE
LT FAIL 2	94	I	LT FAIL 2
DTSTR	95	I	DTSTR
ADPTR BREQ	4	O	SS ADPTR BUS REQ
RMT PARITY 4	12	O	RMT TERM ADRS PARITY 4
RMT PARITY 3	13	O	RMT TERM ADRS PARITY 3
RMT PARITY 2	14	O	RMT TERM ADRS PARITY 2
RMT PARITY 1	15	O	RMT TERM ADRS PARITY 1
RMT PARITY 0	16	O	RMT TERM ADRS PARITY 0
CYCLE ENBL	21	O	CYCLE ENBL
BG/DTGRT	24	O	BG [BC]/DTGRT [RT]
RMT T/BUSCON	28	O	RMT TERM/[L] BUS CONT
SS FLAG	29	O	SS FLAG
ADBC	30	O	ADBC [RMT TERM]
SRQ	32	O	SRQ [RMT TERM]
BUSY	33	O	BUSY [RMT TERM]
CH PRES OUT	38	O	CHAN PRES OUT

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
LOOPBACK	58	O	LOOPBACK
RMT PARITY	76	O	RMT TERM ADRS PARITY
MAE ADRS 7	91	O	MAE ADRS 7
MAE ADRS 6	92	O	MAE ADRS 6
MAE ADRS 5	93	O	MAE ADRS 5
MAE ADRS 4	96	O	MAE ADRS 4
MAE ADRS 3	97	O	MAE ADRS 3
MAE ADRS 2	98	O	MAE ADRS 2
MAE ADRS 1	99	O	MAE ADRS 1
MAE ADRS 0	100	O	MAE ADRS 0

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0264)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation	C	Frame	10	INF	MOhm	1-8

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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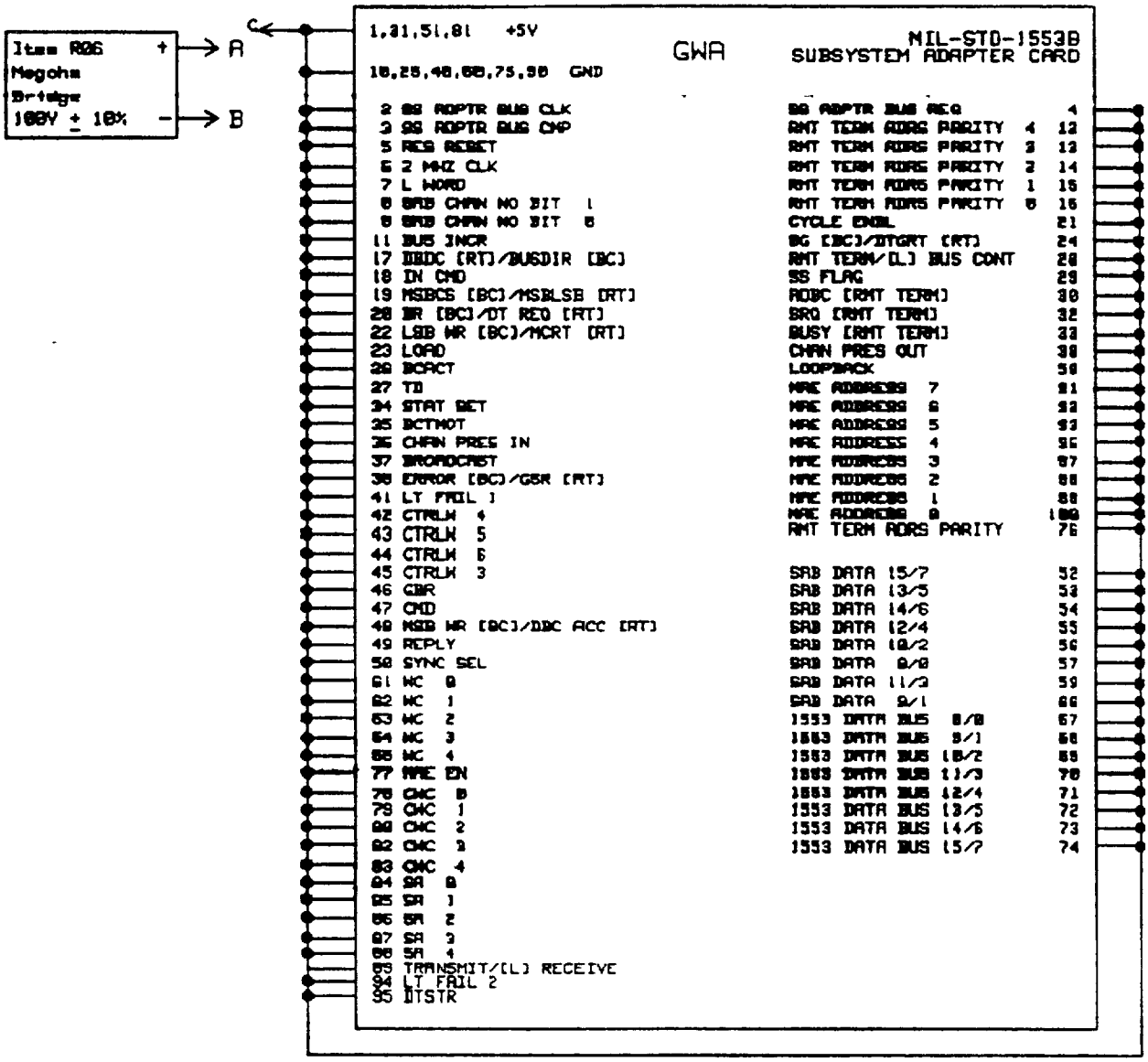


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,31,51,81 +5V

10,25,49,60,75,90 GND

GWA

MIL-STD-1553B
SUBSYSTEM ADAPTER CARD

2	SS RDPTR BUS CLK	SS RDPTR BUS REQ	4
3	SS RDPTR BUS CMP	RMT TERM RDRS PARITY 4	12
6	RES RESET	RMT TERM RDRS PARITY 3	13
5	2 MHZ CLK	RMT TERM RDRS PARITY 2	14
7	L WORD	RMT TERM RDRS PARITY 1	15
8	ERR CHAN NO BIT 1	RMT TERM RDRS PARITY 0	16
9	ERR CHAN NO BIT 0	CYCLE ENBL	21
11	BUS INCR	BG [BC]/DTGRT [RT]	24
17	DRDC [RT]/BUSDIR [BC]	RMT TERM/[L] BUS CONT	28
18	IN CMD	SS FLAG	29
19	MSBCS [BC]/MSBLSB [RT]	ADDC [RMT TERM]	30
20	BR [BC]/DT REQ [RT]	SRQ [RMT TERM]	32
22	LSB WR [BC]/MCRT [RT]	BUSY [RMT TERM]	33
23	LORD	CHAN PRES OUT	38
26	BCRCT	LOOPBACK	50
27	TD	MAE ADDRESS 7	91
34	STRT SET	MAE ADDRESS 6	92
35	BCTMOT	MAE ADDRESS 5	93
36	CHAN PRES IN	MAE ADDRESS 4	96
37	BROADCRST	MAE ADDRESS 3	97
39	ERROR [BC]/SSR [RT]	MAE ADDRESS 2	98
41	LT FAIL 1	MAE ADDRESS 1	99
42	CTRLW 4	MAE ADDRESS 0	100
43	CTRLW 5	RMT TERM RDRS PARITY	76
44	CTRLW 6		
45	CTRLW 3	SAB DATA 15/7	52
46	GBR	SAB DATA 13/5	53
47	CMD	SAB DATA 14/6	54
48	MSB WR [BC]/DBC ACC [RT]	SAB DATA 12/4	55
49	REPLY	SAB DATA 10/2	56
50	SYNC SEL	SAB DATA 8/0	57
51	MC 0	SAB DATA 11/3	58
52	MC 1	SAB DATA 9/1	66
53	MC 2	1553 DATA BUS 8/0	67
54	MC 3	1553 DATA BUS 9/1	68
55	MC 4	1553 DATA BUS 10/2	69
77	MAE EN	1553 DATA BUS 11/3	70
78	CMC 0	1553 DATA BUS 12/4	71
79	CMC 1	1553 DATA BUS 13/5	72
80	CMC 2	1553 DATA BUS 14/6	73
82	CMC 3	1553 DATA BUS 15/7	74
83	CMC 4		
84	SA 0		
85	SA 1		
86	SA 2		
87	SA 3		
88	SA 4		
89	TRANSMIT/[L] RECEIVE		
94	LT FAIL 2		
95	DTSTR		

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	3.96 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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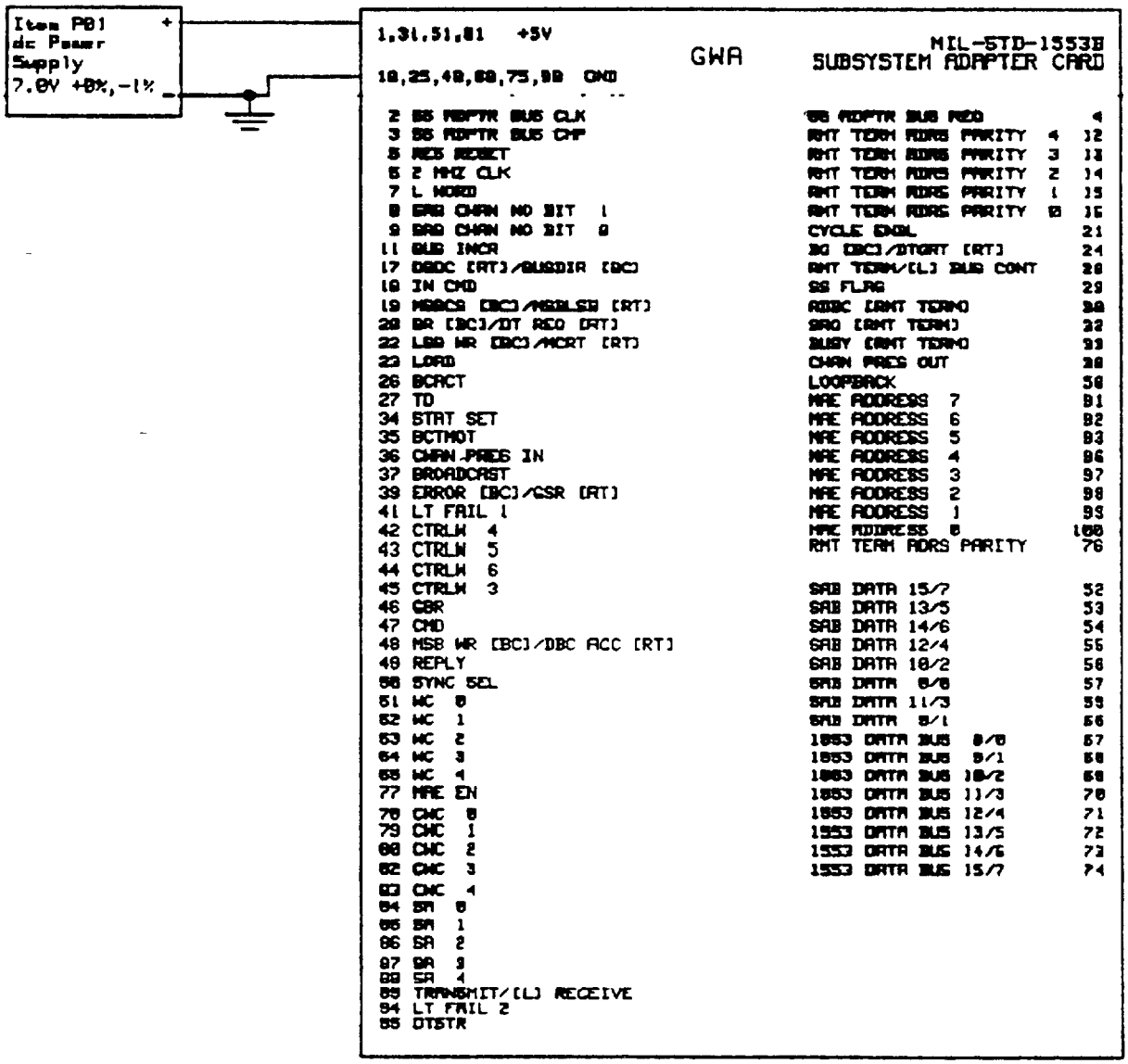


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	ADPTR BCLK	2	5.5	0.0	0.5	mA	1-3
D2	ADPTR BCMP	3	5.5	0.0	1.1	mA	1-3
D3	RES RESET	5	5.5	0.0	1.0	mA	1-3
D4	2 MHZ CLK	6					
D5	L WORD	7	5.5	0.0	1.0	mA	1-3
D6	SAB CHAN B1	8	5.5	0.0	1.2	mA	1-3
D7	SAB CHAN B0	9	5.5	0.0	1.2	mA	1-3
D8	BUS INCR	11	5.5	0.0	0.1	mA	1-3
D9	DBDC/BUSDIR	17	5.5	0.0	2.0	mA	1-3
D10	IN CMD	18	5.5	0.0	1.0	mA	1-3
D11	MSBCS/MSBLSB	19	5.5	0.0	3.0	mA	1-3
D12	BR/DT REQ	20	5.5	0.0	1.0	mA	1-3
D13	LSB WR/MCRT	22	5.5	-0.024	2.02	mA	1-3
D14	LOAD	23	5.5	0.0	2.1	mA	1-3
D15	BCACT	26	5.5	0.0	4.0	mA	1-3
D16	TD	27	5.5	0.0	2.0	mA	1-3
D17	STATSET	34	5.5	0.0	2.2	mA	1-3
D18	BCTMOT	35	5.5	0.0	2.1	mA	1-3
D19	CH PRES IN	36	5.5	0.0	1.0	mA	1-3
D20	BROADCAST	37	5.5	0.0	1.0	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	-Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	ERROR/GSR	39	5.5	0.0	2.0	mA	1-3
D22	LT FAIL 1	41	5.5	0.0	1.0	mA	1-3
D23	CTRLW 4	42					
D24	CTRLW 5	43					
D25	CTRLW 6	44	5.5	0.0	1.0	mA	1-3
D26	CTRLW 3	45	5.5	0.0	2.0	mA	1-3
D27	GBR	46	5.5	0.0	1.0	mA	1-3
D28	CMD	47	5.5	0.0	2.0	mA	1-3
D29	MSBWR/DBCACC	48	5.5	-0.024	3.02	mA	1-3
D30	REPLY	49	5.5	0.0	2.0	mA	1-3
D31	SYNC SEL	50	5.5	0.0	1.0	mA	1-3
D32	WC 0	61	5.5	0.0	0.005	mA	1-3
D33	WC 1	62					
D34	WC 2	63					
D35	WC 3	64					
D36	WC 4	65	5.5	0.0	0.005	mA	1-3
D37	MAE EN	77	5.5	0.0	0.1	mA	1-3
D38	CWC 0	78	5.5	0.0	0.005	mA	1-3
D39	CWC 1	79					
D40	CWC 2	80	5.5	0.0	0.005	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	CWC 3	82	5.5	0.0	0.005	mA	1-3
D42	CWC 4	83	5.5	0.0	0.005	mA	1-3
D43	SA 0	84	5.5	0.0	0.01	mA	1-3
D44	SA 1	85					
D45	SA 2	86	5.5	0.0	0.01	mA	1-3
D46	SA 3	87	5.5	0.0	0.005	mA	1-3
D47	SA 4	88	5.5	0.0	0.005	mA	1-3
D48	TRANS/REC	89	5.5	0.0	2.005	mA	1-3
D49	LT FAIL 2	94	5.5	0.0	1.0	mA	1-3
D50	DTSTR	95	5.5	0.0	1.0	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D50
Power supply current limit	3.3 A	NA	D1-D50
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D50
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D50 D1-D50
Input current measurement	NA	± 25.0 uA	D1
	NA	± 55.0 uA	D2
	NA	± 50.0 uA	D3-D5
	NA	± 60.0 uA	D6,D7
	NA	± 5.0 uA	D8
	NA	± 100.0 uA	D9
	NA	± 50.0 uA	D10
	NA	± 150.0 uA	D11
	NA	± 50.0 uA	D12
	NA	± 102.0 uA	D13
	NA	± 105.0 uA	D14
	NA	± 200.0 uA	D15
	NA	± 100.0 uA	D16
	NA	± 110.0 uA	D17
	NA	± 105.0 uA	D18
	NA	± 50.0 uA	D19,D20
	NA	± 100.0 uA	D21
	NA	± 50.0 uA	D22-D25
	NA	± 100.0 uA	D26
	NA	± 50.0 uA	D27
	NA	± 100.0 uA	D28
	NA	± 152.0 uA	D29
	NA	± 100.0 uA	D30

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 50.0 uA	D31
	NA	± 0.25 uA	D32-D36
	NA	± 5.0 uA	D37
	NA	± 0.25 uA	D38-D42
	NA	± 0.5 uA	D43-D45
	NA	± 0.25 uA	D46,D47
	NA	± 100.0 uA	D48
	NA	± 50.0 uA	D49,D50

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	-25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
E1	ADPTR BREQ	4	0.0	-250.0	-30.0	mA	1-4
E2	RMT PARITY 4	12	0.0	-350.0	-60.0	mA	1-4
E3	RMT PARITY 3	13					
E4	RMT PARITY 2	14					
E5	RMT PARITY 1	15					
E6	RMT PARITY 0	16	0.0	-350.0	-60.0	mA	1-4
E7	CYCLE ENBL	21	0.0	-250.0	-30.0	mA	1-4
E8	BG/DTGRT	24	0.0	-250.0	-30.0	mA	1-4
E9	RMT T/BUSCON	28	0.0	-350.0	-60.0	mA	1-4
E10	SS FLAG	29	0.0	-150.0	-60.0	mA	1-4
E11	ADBC	30					
E12	SRQ	32	0.0	-150.0	-60.0	mA	1-4
E13	BUSY	33	0.0	-220.0	-60.0	mA	1-4
E14	SABDATA 15/7	52	0.0	-350.0	-60.0	mA	1-4
E15	SABDATA 13/5	53					
E16	SABDATA 14/6	54					
E17	SABDATA 12/4	55					
E18	SABDATA 10/2	56					
E19	SABDATA 8/0	57					
E20	LOOPBACK	58	0.0	-350.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	SABDATA 11/3	59	0.0	-350.0	-60.0	mA	1-4
E22	SABDATA 9/1	66					
E23	1553BUS 8/0	67					
E24	1553BUS 8/0	67					
E25	1553BUS 9/1	68					
E26	1553BUS 9/1	68					
E27	1553BUS 10/2	69					
E28	1553BUS 10/2	69					
E29	1553BUS 11/3	70					
E30	1553BUS 11/3	70					
E31	1553BUS 12/4	71					
E32	1553BUS 12/4	71					
E33	1553BUS 13/5	72					
E34	1553BUS 13/5	72					
E35	1553BUS 14/6	73					
E36	1553BUS 14/6	73					
E37	1553BUS 15/7	74					
E38	1553BUS 15/7	74					
E39	RMT PARITY	76					
E40	MAE ADRS 7	91	0.0	-350.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E41	MAE ADRS 6	92	0.0	-350.0	-60.0	mA	1-4
E42	MAE ADRS 5	93					
E43	MAE ADRS 4	96					
E44	MAE ADRS 3	97					
E45	MAE ADRS 2	98					
E46	MAE ADRS 1	99					
E47	MAE ADRS 0	100	0.0	-350.0	-60.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E47
Power supply current limit	3.3 A	NA	E1-E47
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E47
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E47 E1-E47
Output current measurement	NA NA NA NA NA NA NA	± 11.0 mA ± 14.5 mA ± 11.0 mA ± 14.5 mA ± 4.5 mA ± 8.0 mA ± 14.5 mA	E1 E2-E6 E7,E8 E9 E10-E12 E13 E14-E47

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	ADPTR BCLK	2	2.4	-3.0	0.1	mA	1-3
F2	ADPTR BCMP	3	2.4	-0.85	0.045	mA	1-3
F3	RES RESET	5	2.4	-0.25	0.025	mA	1-3
F4	2 MHZ CLK	6					
F5	L WORD	7	2.4	-0.25	0.025	mA	1-3
F6	SAB CHAN B1	8	2.4	-0.855	0.05	mA	1-3
F7	SAB CHAN B0	9	2.4	-0.855	0.05	mA	1-3
F8	BUS INCR	11	2.4	-0.6	0.02	mA	1-3
F9	DBDC/BUSDIR	17	2.4	-0.5	0.05	mA	1-3
F10	IN CMD	18	2.4	-0.25	0.025	mA	1-3
F11	MSBCS/MSBLSB	19	2.4	-0.75	0.075	mA	1-3
F12	BR/DT REQ	20	2.4	-0.25	0.025	mA	1-3
F13	LSB WR/MCRT	22	2.4	-1.65	-0.562	mA	1-3
F14	LOAD	23	2.4	-0.7	0.07	mA	1-3
F15	BCACT	26	2.4	-1.0	0.1	mA	1-3
F16	TD	27	2.4	-0.5	0.05	mA	1-3
F17	STATSET	34	2.4	-1.1	0.09	mA	1-3
F18	BCTMOT	35	2.4	-0.9	0.07	mA	1-3
F19	CH PRES IN	36	2.4	-0.25	0.025	mA	1-3
F20	BROADCAST	37	2.4	-0.25	0.025	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	ERROR/GSR	39	2.4	-0.5	0.05	mA	1-3
F22	LT FAIL 1	41	2.4	-0.25	0.025	mA	1-3
F23	CTRLW 4	42					
F24	CTRLW 5	43					
F25	CTRLW 6	44	2.4	-0.25	0.025	mA	1-3
F26	CTRLW 3	45	2.4	-0.5	0.05	mA	1-3
F27	GBR	46	2.4	-0.25	0.025	mA	1-3
F28	CMD	47	2.4	-0.5	0.05	mA	1-3
F29	MSBWR/DBCACC	48	2.4	-1.9	-0.537	mA	1-3
F30	REPLY	49	2.4	-0.5	0.05	mA	1-3
F31	SYNC SEL	50	2.4	-0.25	0.025	mA	1-3
F32	SABDATA 15/7	52	2.4	-0.02	0.02	mA	1-3
F33	SABDATA 13/5	53					
F34	SABDATA 14/6	54					
F35	SABDATA 12/4	55					
F36	SABDATA 10/2	56					
F37	SABDATA 8/0	57					
F38	SABDATA 11/3	59	2.4	-0.02	0.02	mA	1-3
F39	WC 0	61	2.4	-0.005	0.005	mA	1-3
F40	WC 1	62	2.4	-0.005	0.005	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	WC 2	63	2.4	-0.005	0.005	mA	1-3
F42	WC 3	64					
F43	WC 4	65	2.4	-0.005	0.005	mA	1-3
F44	SABDATA 9/1	66	2.4	-0.02	0.02	mA	1-3
F45	1553BUS 8/0	67	2.4	-0.23	0.05	mA	1-3
F46	1553BUS 9/1	68	2.4	-0.03	0.03	mA	1-3
F47	1553BUS 10/2	69	2.4	-0.03	0.03	mA	1-3
F48	1553BUS 11/3	70	2.4	-0.23	0.05	mA	1-3
F49	1553BUS 12/4	71	2.4	-0.03	0.03	mA	1-3
F50	1553BUS 13/5	72					
F51	1553BUS 14/6	73					
F52	1553BUS 15/7	74	2.4	-0.03	0.03	mA	1-3
F53	MAE EN	77	2.4	-0.6	0.02	mA	1-3
F54	CWC 0	78	2.4	-0.005	0.005	mA	1-3
F55	CWC 1	79					
F56	CWC 2	80					
F57	CWC 3	82					
F58	CWC 4	83	2.4	-0.005	0.005	mA	1-3
F59	SA 0	84	2.4	-0.01	0.01	mA	1-3
F60	SA 1	85	2.4	-0.01	0.01	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F61	SA 2	86	2.4	-0.01	0.01	mA	1-3
F62	SA 3	87	2.4	-0.005	0.005	mA	1-3
F63	SA 4	88	2.4	-0.005	0.005	mA	1-3
F64	TRANS/REC	89	2.4	-0.5	0.05	mA	1-3
F65	LT FAIL 2	94	2.4	-0.25	0.025	mA	1-3
F66	DTSTR	95	2.4	-0.25	0.025	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F66
Power supply current limit	3.3 A	NA	F1-F66
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F66
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F66
low, (VL)	0.0 V	± 100 mV	F1-F66
Input current measurement	NA	± 155.0 uA	F1
	NA	± 44.7 uA	F2
	NA	± 13.7 uA	F3-F5
	NA	± 45.2 uA	F6,F7
	NA	± 31.0 uA	F8
	NA	± 27.5 uA	F9
	NA	± 13.7 uA	F10
	NA	± 41.2 uA	F11
	NA	± 13.7 uA	F12
	NA	± 54.4 uA	F13
	NA	± 38.5 uA	F14
	NA	± 55.0 uA	F15
	NA	± 27.5 uA	F16
	NA	± 59.5 uA	F17
	NA	± 48.5 uA	F18
	NA	± 13.7 uA	F19,F20
	NA	± 27.5 uA	F21
	NA	± 13.7 uA	F22-F25
	NA	± 27.5 uA	F26
	NA	± 13.7 uA	F27
	NA	± 27.5 uA	F28
	NA	± 68.1 uA	F29

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 27.5 uA	F30
	NA	± 13.7 uA	F31
	NA	± 2.0 uA	F32-F38
	NA	± 0.5 uA	F39-F43
	NA	± 2.0 uA	F44
	NA	± 14.0 uA	F45
	NA	± 3.0 uA	F46,F47
	NA	± 14.0 uA	F48
	NA	± 3.0 uA	F49-F52
	NA	± 31.0 uA	F53
	NA	± 0.5 uA	F54-F58
	NA	± 1.0 uA	F59-F61
	NA	± 0.5 uA	F62,F63
	NA	± 27.5 uA	F64
	NA	± 13.7 uA	F65,F66

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	ADPTR BCLK	2	0.4	-3.0	0.1	mA	1-3
G2	ADPTR BCMP	3	0.4	-0.85	0.045	mA	1-3
G3	RES RESET	5	0.4	-0.25	0.025	mA	1-3
G4	2 MHZ CLK	6					
G5	L WORD	7	0.4	-0.25	0.025	mA	1-3
G6	SAB CHAN B1	8	0.4	-0.855	0.05	mA	1-3
G7	SAB CHAN B0	9	0.4	-0.855	0.05	mA	1-3
G8	BUS INCR	11	0.4	-0.6	0.02	mA	1-3
G9	DBDC/BUSDIR	17	0.4	-0.5	0.05	mA	1-3
G10	IN CMD	18	0.4	-0.25	0.025	mA	1-3
G11	MSBCS/MSBLSB	19	0.4	-0.75	0.075	mA	1-3
G12	BR/DT REQ	20	0.4	-0.25	0.025	mA	1-3
G13	LSB WR/MCRT	22	0.4	-1.65	-0.562	mA	1-3
G14	LOAD	23	0.4	-0.7	0.07	mA	1-3
G15	BCACT	26	0.4	-1.0	0.1	mA	1-3
G16	TD	27	0.4	-0.5	0.05	mA	1-3
G17	STATSET	34	0.4	-1.1	0.09	mA	1-3
G18	BCTMOT	35	0.4	-0.9	0.07	mA	1-3
G19	CH PRES IN	36	0.4	-0.25	0.025	mA	1-3
G20	BROADCAST	37	0.4	-0.25	0.025	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	ERROR/GSR	39	0.4	-0.5	0.05	mA	1-3
G22	LT FAIL 1	41	0.4	-0.25	0.025	mA	1-3
G23	CTRLW 4	42					
G24	CTRLW 5	43					
G25	CTRLW 6	44	0.4	-0.25	0.025	mA	1-3
G26	CTRLW 3	45	0.4	-0.5	0.05	mA	1-3
G27	GBR	46	0.4	-0.25	0.025	mA	1-3
G28	CMD	47	0.4	-0.5	0.05	mA	1-3
G29	MSBWR/DBCACC	48	0.4	-1.9	-0.537	mA	1-3
G30	REPLY	49	0.4	-0.5	0.05	mA	1-3
G31	SYNC SEL	50	0.4	-0.25	0.025	mA	1-3
G32	SABDATA 15/7	52	0.4	-0.02	0.02	mA	1-3
G33	SABDATA 13/5	53					
G34	SABDATA 14/6	54					
G35	SABDATA 12/4	55					
G36	SABDATA 10/2	56					
G37	SABDATA 8/0	57					
G38	SABDATA 11/3	59	0.4	-0.02	0.02	mA	1-3
G39	WC 0	61	0.4	-0.005	0.005	mA	1-3
G40	WC 1	62	0.4	-0.005	0.005	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	WC 2	63	0.4	-0.005	0.005	mA	1-3
G42	WC 3	64					
G43	WC 4	65	0.4	-0.005	0.005	mA	1-3
G44	SABDATA 9/1	66	0.4	-0.02	0.02	mA	1-3
G45	1553BUS 8/0	67	0.4	-0.23	0.05	mA	1-3
G46	1553BUS 9/1	68	0.4	-0.03	0.03	mA	1-3
G47	1553BUS 10/2	69	0.4	-0.03	0.03	mA	1-3
G48	1553BUS 11/3	70	0.4	-0.23	0.05	mA	1-3
G49	1553BUS 12/4	71	0.4	-0.03	0.03	mA	1-3
G50	1553BUS 13/5	72					
G51	1553BUS 14/6	73					
G52	1553BUS 15/7	74	0.4	-0.03	0.03	mA	1-3
G53	MAE EN	77	0.4	-0.6	0.02	mA	1-3
G54	CWC 0	78	0.4	-0.005	0.005	mA	1-3
G55	CWC 1	79					
G56	CWC 2	80					
G57	CWC 3	82					
G58	CWC 4	83	0.4	-0.005	0.005	mA	1-3
G59	SA 0	84	0.4	-0.01	0.01	mA	1-3
G60	SA 1	85	0.4	-0.01	0.01	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G61	SA 2	86	0.4	-0.01	0.01	mA	1-3
G62	SA 3	87	0.4	-0.005	0.005	mA	1-3
G63	SA 4	88	0.4	-0.005	0.005	mA	1-3
G64	TRANS/REC	89	0.4	-0.5	0.05	mA	1-3
G65	LT FAIL 2	94	0.4	-0.25	0.025	mA	1-3
G66	DTSTR	95	0.4	-0.25	0.025	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:13Z
Check the source to verify that this is the current version before use.

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G66
Power supply current limit	3.3 A	NA	G1-G66
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G66
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	G1-G66
low, (VL)	0.0 V	± 100 mV	G1-G66
Input current measurement	NA	± 155.0 uA	G1
	NA	± 44.7 uA	G2
	NA	± 13.7 uA	G3-G5
	NA	± 45.2 uA	G6,G7
	NA	± 31.0 uA	G8
	NA	± 27.5 uA	G9
	NA	± 13.7 uA	G10
	NA	± 41.2 uA	G11
	NA	± 13.7 uA	G12
	NA	± 54.4 uA	G13
	NA	± 38.5 uA	G14
	NA	± 55.0 uA	G15
	NA	± 27.5 uA	G16
	NA	± 59.5 uA	G17
	NA	± 48.5 uA	G18
	NA	± 13.7 uA	G19,G20
	NA	± 27.5 uA	G21
	NA	± 13.7 uA	G22-G25
	NA	± 27.5 uA	G26
	NA	± 13.7 uA	G27
	NA	± 27.5 uA	G28
	NA	± 68.1 uA	G29
	NA	± 27.5 uA	G30

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 13.7 uA	G31
	NA	± 2.0 uA	G32-G38
	NA	± 0.5 uA	G39-G43
	NA	± 2.0 uA	G44
	NA	± 14.0 uA	G45
	NA	± 3.0 uA	G46,G47
	NA	± 14.0 uA	G48
	NA	± 3.0 uA	G49-G52
	NA	± 31.0 uA	G53
	NA	± 0.5 uA	G54-G58
	NA	± 1.0 uA	G59-G61
	NA	± 0.5 uA	G62,G63
	NA	± 27.5 uA	G64
	NA	± 13.7 uA	G65,G66

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	ADPTR BREQ	4	-0.16	2.4	4.5	V	1-4
H2	RMT PARITY 4	12	-0.4				
H3	RMT PARITY 3	13					
H4	RMT PARITY 2	14					
H5	RMT PARITY 1	15					
H6	RMT PARITY 0	16	-0.4				
H7	CYCLE ENBL	21	-0.16				
H8	BG/DTGRT	24					
H9	RMT T/BUSCON	28	-0.16				
H10	SS FLAG	29	-0.4				
H11	ADBC	30					
H12	SRQ	32	-0.4				
H13	BUSY	33	-0.16				
H14	SABDATA 15/7	52	-0.4				
H15	SABDATA 13/5	53					
H16	SABDATA 14/6	54					
H17	SABDATA 12/4	55					
H18	SABDATA 10/2	56					
H19	SABDATA 8/0	57					
H20	LOOPBACK	58	-0.4	2.4	4.5	V	1-4

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Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	SABDATA 11/3	59	-0.4	2.4	4.5	V	1-4
H22	SABDATA 9/1	66					
H23	1553BUS 8/0	67					
H24	1553BUS 8/0	67					
H25	1553BUS 9/1	68					
H26	1553BUS 9/1	68					
H27	1553BUS 10/2	69					
H28	1553BUS 10/2	69					
H29	1553BUS 11/3	70					
H30	1553BUS 11/3	70					
H31	1553BUS 12/4	71					
H32	1553BUS 12/4	71					
H33	1553BUS 13/5	72					
H34	1553BUS 13/5	72					
H35	1553BUS 14/6	73					
H36	1553BUS 14/6	73					
H37	1553BUS 15/7	74					
H38	1553BUS 15/7	74					
H39	RMT PARITY	76					
H40	MAE ADRS 7	91	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H41	MAE ADRS 6	92	-0.4	2.4	4.5	V	1-4
H42	MAE ADRS 5	93					
H43	MAE ADRS 4	96					
H44	MAE ADRS 3	97					
H45	MAE ADRS 2	98					
H46	MAE ADRS 1	99					
H47	MAE ADRS 0	100	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H47
Power supply current limit	3.3 A	NA	H1-H47
Forcing current, (Ioh)	-0.16 mA -0.4 mA -0.16 mA -0.4 mA -0.16 mA -0.4 mA	± 16 uA ± 40 uA ± 16 uA ± 40 uA ± 16 uA ± 40 uA	H1 H2-H6 H7-H9 H10-H12 H13 H14-H47
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1-H47 H1-H47
Output voltage measurement	NA	± 105 mV	H1-H47

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

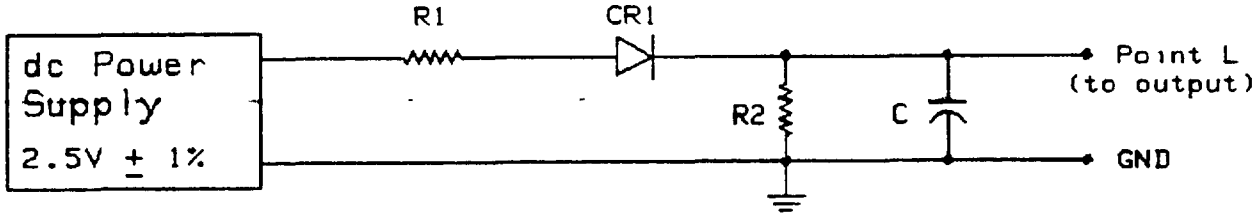
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
4,21,24,28,33	16000	1.6	0.5
38	NA	1.6	0.5
12,13,14,15,16, 29,30,32,52,53, 54,55,56,57,58, 59,66,67,68,69, 70,71,72,73,74, 76,91,92,93,96, 97,98,99,100	6200	6.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	ADPTR BREQ	4	1.6	0.0	0.5	V	1-4
J2	RMT PARITY 4	12	6.0				
J3	RMT PARITY 3	13					
J4	RMT PARITY 2	14					
J5	RMT PARITY 1	15					
J6	RMT PARITY 0	16	6.0				
J7	CYCLE ENBL	21	1.6				
J8	BG/DTGRT	24					
J9	RMT T/BUSCON	28	1.6				
J10	SS FLAG	29	6.0				
J11	ADBC	30					
J12	SRQ	32	6.0				
J13	BUSY	33	1.6				
J14	CH PRES OUT	38	1.6				
J15	SABDATA 15/7	52	6.0				
J16	SABDATA 13/5	53					
J17	SABDATA 14/6	54					
J18	SABDATA 12/4	55					
J19	SABDATA 10/2	56					
J20	SABDATA 8/0	57	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	LOOPBACK	58	6.0	0.0	0.5	V	1-4
J22	SABDATA 11/3	59					
J23	SABDATA 9/1	66					
J24	1553BUS 8/0	67					
J25	1553BUS 8/0	67					
J26	1553BUS 9/1	68					
J27	1553BUS 9/1	68					
J28	1553BUS 10/2	69					
J29	1553BUS 10/2	69					
J30	1553BUS 11/3	70					
J31	1553BUS 11/3	70					
J32	1553BUS 12/4	71					
J33	1553BUS 12/4	71					
J34	1553BUS 13/5	72					
J35	1553BUS 13/5	72					
J36	1553BUS 14/6	73					
J37	1553BUS 14/6	73					
J38	1553BUS 15/7	74					
J39	1553BUS 15/7	74					
J40	RMT PARITY	76	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J41	MAE ADRS 7	91	6.0	0.0	0.5	V	1-4
J42	MAE ADRS 6	92					
J43	MAE ADRS 5	93					
J44	MAE ADRS 4	96					
J45	MAE ADRS 3	97					
J46	MAE ADRS 2	98					
J47	MAE ADRS 1	99					
J48	MAE ADRS 0	100	6.0	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:13Z
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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J48
Power supply current limit	3.3 A	NA	J1-J48
Forcing current, (Iol)	1.6 mA 6.0 mA 1.6 mA 6.0 mA 1.6 mA 6.0 mA	± 160 uA ± 600 uA ± 160 uA ± 600 uA ± 160 uA ± 600 uA	J1 J2-J6 J7-J9 J10-J12 J13,J14 J15-J48
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J48 J1-J48
Output voltage measurement	NA	± 25 mV	J1-J48

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
High level							
K1	MAE ADRS 7	91	2.4	-0.01	0.01	mA	1-3
K2	MAE ADRS 6	92					
K3	MAE ADRS 5	93					
K4	MAE ADRS 4	96					
K5	MAE ADRS 3	97					
K6	MAE ADRS 2	98					
K7	MAE ADRS 1	99					
K8	MAE ADRS 0	100	2.4	-0.01	0.01	mA	1-3
Low level							
K9	MAE ADRS 7	91	0.4	-0.01	0.01	mA	1-3
K10	MAE ADRS 6	92					
K11	MAE ADRS 5	93					
K12	MAE ADRS 4	96					
K13	MAE ADRS 3	97					
K14	MAE ADRS 2	98					
K15	MAE ADRS 1	99					
K16	MAE ADRS 0	100	0.4	-0.01	0.01	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K16
Power supply current limit	3.3 A	NA	K1-K16
Forcing voltage, (Voh) (Vol)	 2.4 V 0.4 V	 ± 50 mV ± 50 mV	 K1-K8 K9-K16
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 K1-K8 K9-K16
Output current measurement	NA	± 1 uA	K1-K16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	ADPTR BCLK	2	-18.0	-1.2	0.0	V	1-3
L2	ADPTR BCMP	3	-18.0	-1.2	0.0	V	1-3
L3	RES RESET	5	-18.0	-1.5	0.0	V	1-3
L4	2 MHZ CLK	6					
L5	L WORD	7	-18.0	-1.5	0.0	V	1-3
L6	SAB CHAN B1	8	-18.0	-1.2	0.0	V	1-3
L7	SAB CHAN B0	9					
L8	BUS INCR	11	-18.0	-1.2	0.0	V	1-3
L9	DBDC/BUSDIR	17	-18.0	-1.5	0.0	V	1-3
L10	IN CMD	18					
L11	MSBCS/MSBLSB	19					
L12	BR/DT REQ	20					
L13	LOAD	23					
L14	BCACT	26					
L15	TD	27					
L16	STATSET	34					
L17	BCTMOT	35					
L18	CH PRES IN	36					
L19	BROADCAST	37					
L20	ERROR/GSR	39	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	LT FAIL 1	41	-18.0	-1.5	0.0	V	1-3
L22	CTRLW 4	42					
L23	CTRLW 5	43					
L24	CTRLW 6	44					
L25	CTRLW 3	45					
L26	GBR	46					
L27	CMD	47					
L28	REPLY	49					
L29	SYNC SEL	50	-18.0	-1.5	0.0	V	1-3
L30	WC 0	61	-18.0	-1.2	0.0	V	1-3
L31	WC 1	62					
L32	WC 2	63					
L33	WC 3	64					
L34	WC 4	65					
L35	MAE EN	77					
L36	CWC 0	78					
L37	CWC 1	79					
L38	CWC 2	80					
L39	CWC 3	82					
L40	CWC 4	83	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L41	SA 0	84	-18.0	-1.2	0.0	V	1-3
L42	SA 1	85					
L43	SA 2	86					
L44	SA 3	87					
L45	SA 4	88	-18.0	-1.2	0.0	V	1-3
L46	TRANS/REC	89	-18.0	-1.5	0.0	V	1-3
L47	LT FAIL 2	94					
L48	DTSTR	95	-18.0	-1.5	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L48
Power supply current limit	NA	NA	L1-L48
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L48
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L48 L1-L48
Input voltage measurement	NA NA NA NA NA NA	± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV	L1,L2 L3-L5 L6-L8 L9-L29 L30-L45 L46-L48

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ±5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	11.0	20.5	uF	1-4
M2	PSFC	51	7.5	13.9	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 0.48 uF ± 0.32 uF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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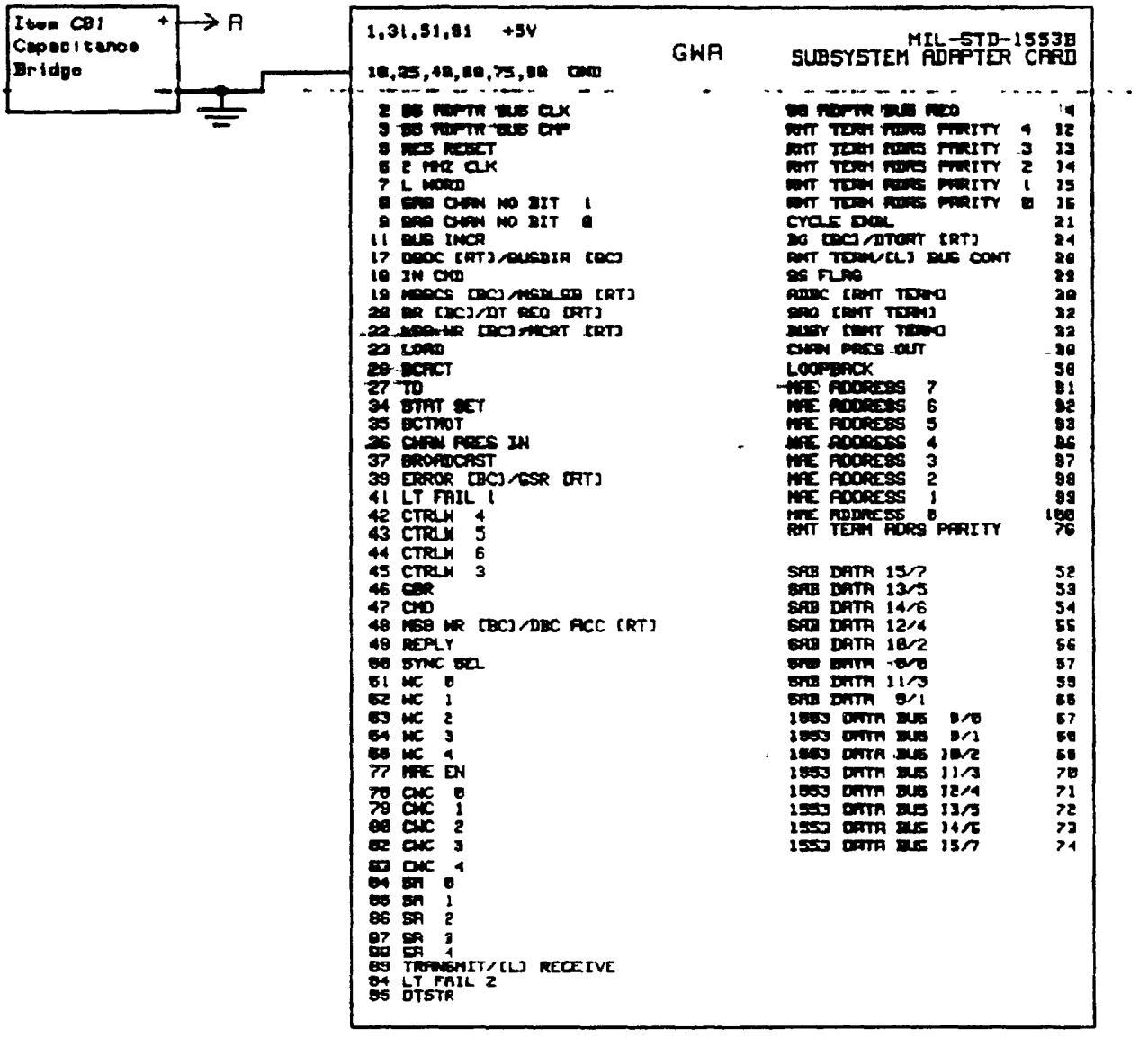


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31 51,81	900	2200	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	3.3 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 65 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	3.3 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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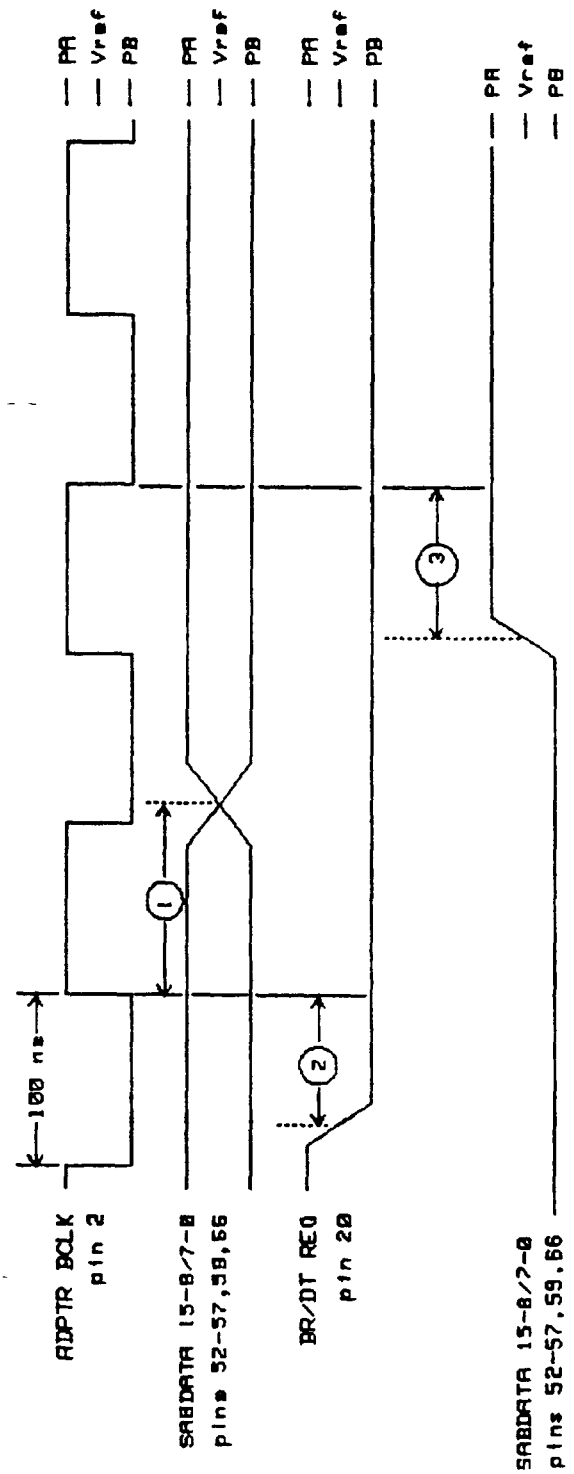


FIGURE 11. Input timing parameters for table XV.

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NOTES:

1. (1) ADPTR BCLK to SABDATA 15-8/7-0.

   Hold time: 100 nS minimum.

2. (2) BR/DT REQ to ADPTR BCLK.

   Setup time: 91 nS minimum.

3. (3) SABDATA 15-8/7-0 to ADPTR BCLK.

   Setup time: 79 nS minimum.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	3.3 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

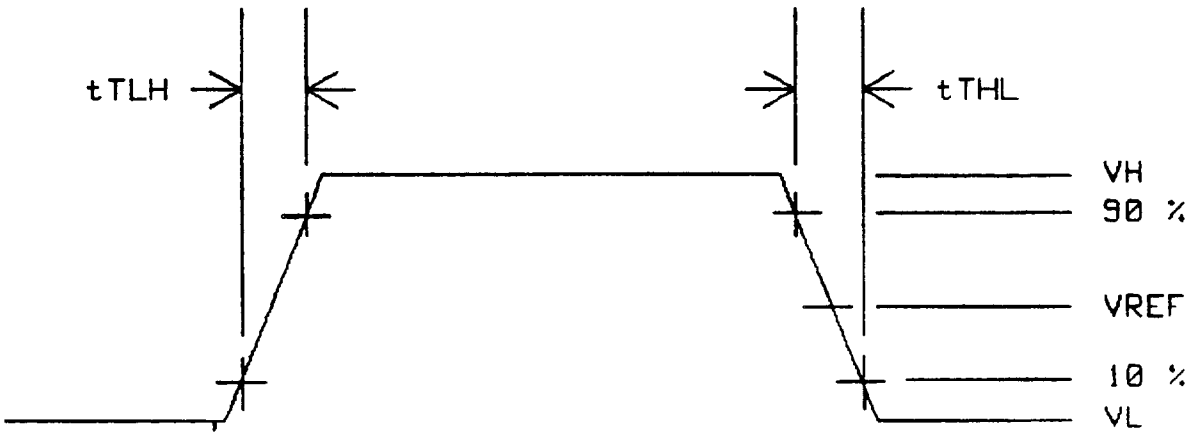
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) $t_{TLH} = t_{THL} \leq 10$ nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	th	ADPTR BCLK to SABDATA 15-8/7-0	100	100	100	-	-	-	nS	12
2	tsu	BR/DT REQ to ADPTR BCLK	91	91	91	-	-	-	nS	12
3	tpd	ADPTR BCLK to ADPTR BREQ	-	-	-	30	30	30	nS	12
4	tpd	ADPTR BCLK to BG/DTGRT	-	-	-	78	78	78	nS	12
5	tpd	ADPTR BCLK to 1553BUS 8-9/0-1	-	-	-	46	46	46	nS	12
5	tpd	ADPTR BCLK to 1553BUS 11-15/3-7	-	-	-	46	46	46	nS	12
6	tpd	LSB WR/MCRT to BUSY	-	-	-	78	78	78	nS	12
7	tpd	MAE EN to MAE ADRS 7-0	-	-	-	18	18	18	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	ADPTR BCLK to SRQ	-	-	-	57	57	57	nS	12
9	tpd	ADPTR BCLK to CYCLE ENBL	-	-	-	45	45	45	nS	12
10	tsu	SABDATA 15-8/7-0 to ADPTR BCLK	79	79	79	-	-	-	nS	12

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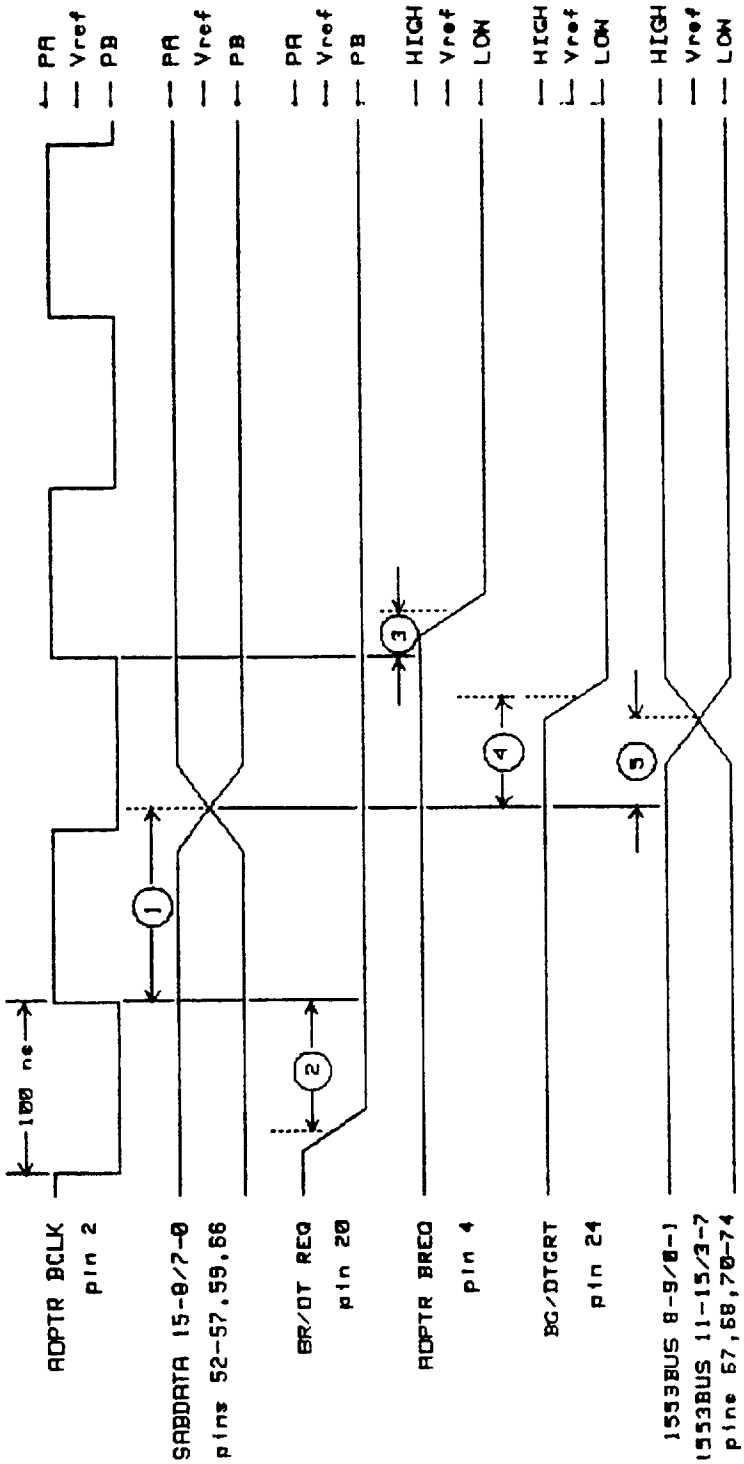


FIGURE 12a. Dynamic waveforms for table XVII.

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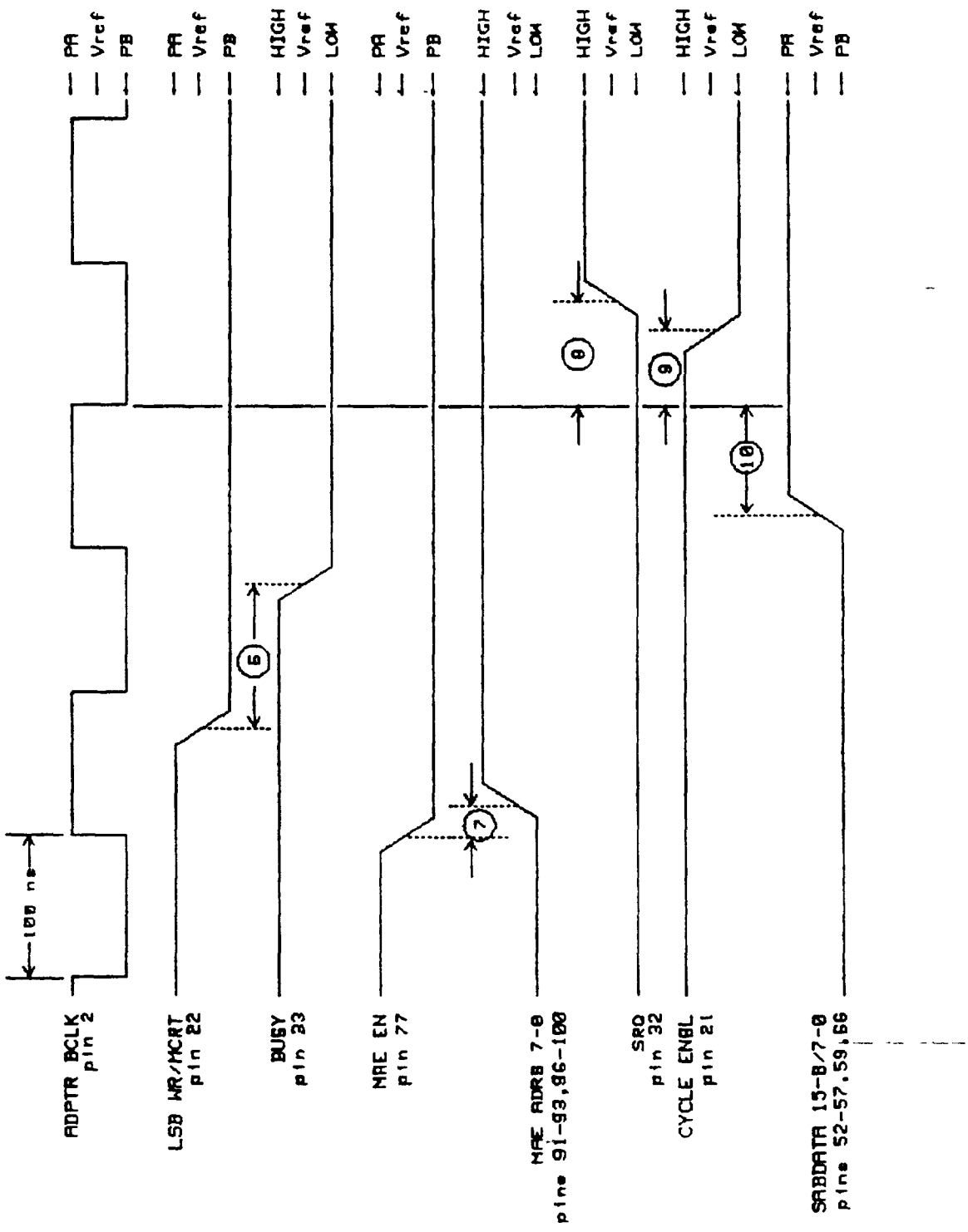


FIGURE 12b. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	ADPTR BCLK	2	8	125	pF	1-4
R2	ADPTR BCMP	3	4	75	pF	1-4
R3	RES RESET	5	2	75	pF	1-4
R4	2 MHZ CLK	6				
R5	L WORD	7	2	75	pF	1-4
R6	SAB CHAN B1	8	4	150	pF	1-4
R7	SAB CHAN B0	9	4	150	pF	1-4
R8	BUS INCR	11	2	75	pF	1-4
R9	DBDC/BUSDIR	17	6	150	pF	1-4
R10	IN CMD	18	2	50	pF	1-4
R11	MSBCS/MSBLSB	19	6	100	pF	1-4
R12	BR/DT REQ	20	2	75	pF	1-4
R13	LSB WR/MCRT	22	4	100	pF	1-4
R14	LOAD	23	6	100	pF	1-4
R15	BCACT	26	4	125	pF	1-4
R16	TD	27	4	75	pF	1-4
R17	STATSET	34	8	125	pF	1-4
R18	BCTMOT	35	6	125	pF	1-4
R19	CH PRES IN	36	2	50	pF	1-4
R20	BROADCAST	37	2	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	ERROR/GSR	39	4	100	pF	1-4
R22	LT FAIL 1	41	2	75	pF	1-4
R23	CTRLW 4	42				
R24	CTRLW 5	43				
R25	CTRLW 6	44	2	75	pF	1-4
R26	CTRLW 3	45	4	75	pF	1-4
R27	GBR	46	2	75	pF	1-4
R28	CMD	47	4	75	pF	1-4
R29	MSBWR/DBCACC	48	6	75	pF	1-4
R30	REPLY	49	4	100	pF	1-4
R31	SYNC SEL	50	2	75	pF	1-4
	Bidirectionals					
R32	SABDATA 15/7	52	6	100	pF	1-4
R33	SABDATA 13/5	53				
R34	SABDATA 14/6	54				
R35	SABDATA 12/4	55				
R36	SABDATA 10/2	56				
R37	SABDATA 8/0	57				
R38	SABDATA 11/3	59	6	100	pF	1-4
	Inputs					
R39	WC 0	61	2	75	pF	1-4
R40	WC 1	62	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	WC 2	63	2	75	pF	1-4
R42	WC 3	64				
R43	WC 4	65	2	75	pF	1-4
	Bidirectionals					
R44	SABDATA 9/1	66	6	100	pF	1-4
R45	1553BUS 8/0	67	2	175	pF	1-4
R46	1553BUS 9/1	68	8	125	pF	1-4
R47	1553BUS 10/2	69	8	100	pF	1-4
R48	1553BUS 11/3	70	2	175	pF	1-4
R49	1553BUS 12/4	71	8	100	pF	1-4
R50	1553BUS 13/5	72				
R51	1553BUS 14/6	73				
R52	1553BUS 15/7	74	8	100	pF	1-4
	Inputs					
R53	MAE EN	77	2	75	pF	1-4
R54	CWC 0	78				
R55	CWC 1	79				
R56	CWC 2	80				
R57	CWC 3	82				
R58	CWC 4	83	2	75	pF	1-4
R59	SA 0	84	4	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R60	SA 1	85	4	75	pF	1-4
R61	SA 2	86	4	75	pF	1-4
R62	SA 3	87	2	75	pF	1-4
R63	SA 4	88	|	|	|	|
R64	TRANS/REC	89	2	75	pF	1-4
R65	DTSTR	95	2	50	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R65
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R65
Capacitance measurement	NA	± 5.85 pF	R1
	NA	± 3.55 pF	R2
	NA	± 3.65 pF	R3-R5
	NA	± 7.3 pF	R6,R7
	NA	± 3.65 pF	R8
	NA	± 7.2 pF	R9
	NA	± 2.4 pF	R10
	NA	± 4.7 pF	R11
	NA	± 3.65 pF	R12
	NA	± 4.8 pF	R13
	NA	± 4.7 pF	R14
	NA	± 6.05 pF	R15
	NA	± 3.55 pF	R16
	NA	± 5.85 pF	R17
	NA	± 5.95 pF	R18
	NA	± 2.4 pF	R19,R20
	NA	± 4.8 pF	R21
	NA	± 3.65 pF	R22-R25
	NA	± 3.55 pF	R26
	NA	± 3.65 pF	R27
	NA	± 3.55 pF	R28
	NA	± 3.45 pF	R29
	NA	± 4.8 pF	R30
	NA	± 3.65 pF	R31
	NA	± 4.7 pF	R32-R38
	NA	± 3.65 pF	R39-R43
	NA	± 4.7 pF	R44
	NA	± 8.65 pF	R45
	NA	± 5.85 pF	R46
	NA	± 4.6 pF	R47
	NA	± 8.65 pF	R48

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 4.6 pF	R49-R52
	NA	± 3.65 pF	R53-R58
	NA	± 3.55 pF	R59-R61
	NA	± 3.65 pF	R62-R64
	NA	± 2.4 pF	R65

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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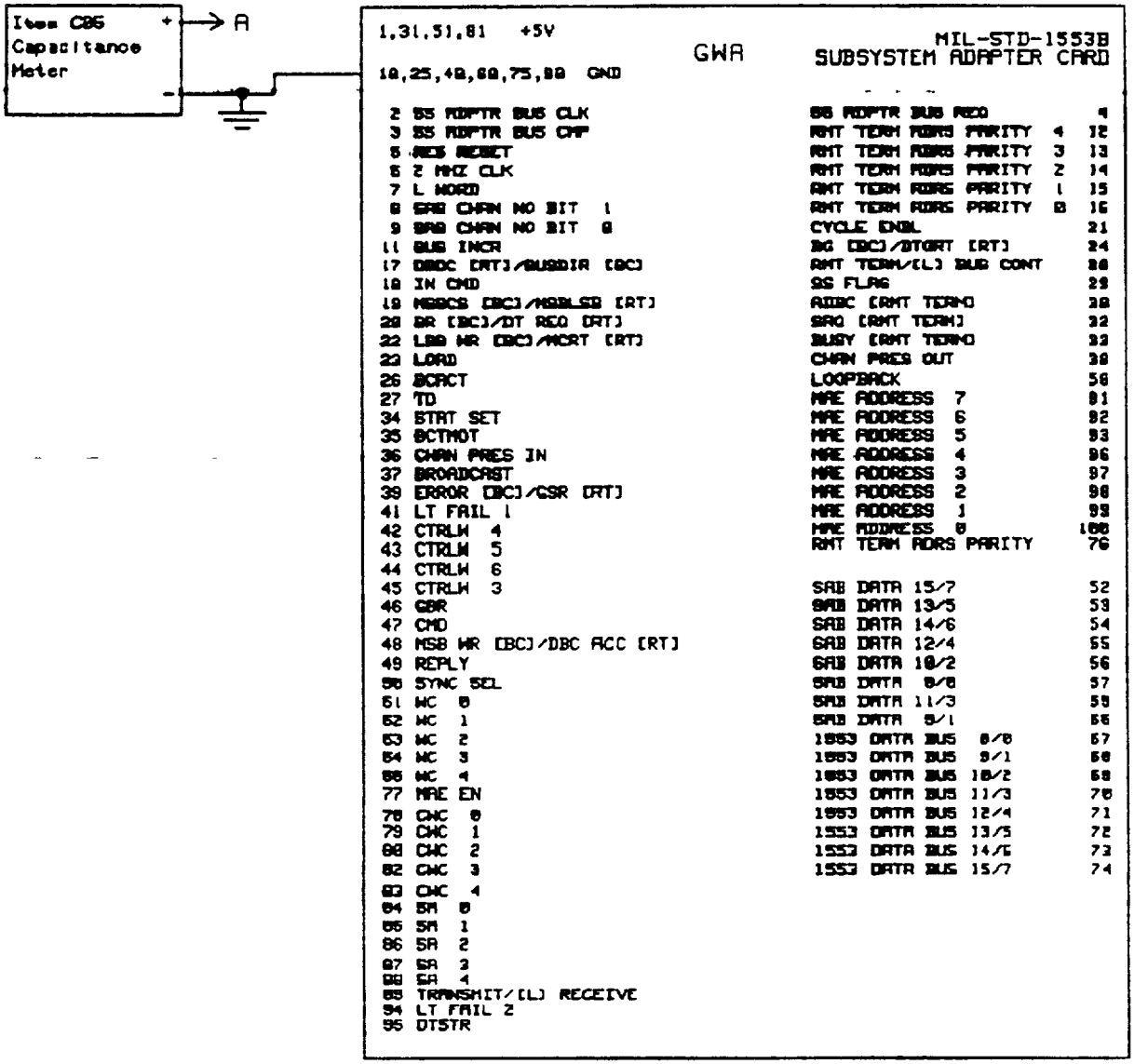


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Input clamp diode latch up test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
S1	ADPTR BCLK	2	-18.0	-1.2	2.4	V	1-3
S2	BUS INCR	11	-18.0	-1.2	2.4	V	1-3

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NOTES:

1. Input clamp diode latch up test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	S1,S2
Power supply current limit	NA	NA	S1,S2
Forcing current, (Iin)	-18 mA	± 1.8 mA	S1,S2
Pattern voltage high, (VH) low, (VL)	NA NA	NA NA	S1,S2 S1,S2
Input voltage measurement	NA	± 180 mV	S1,S2

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XIX and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	51	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	3.3 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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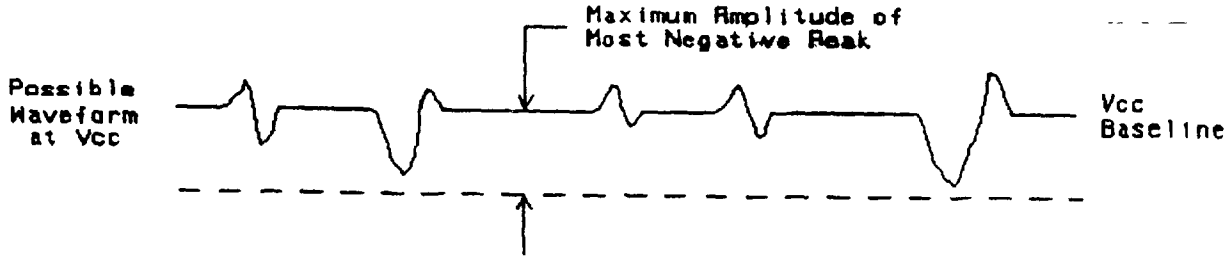
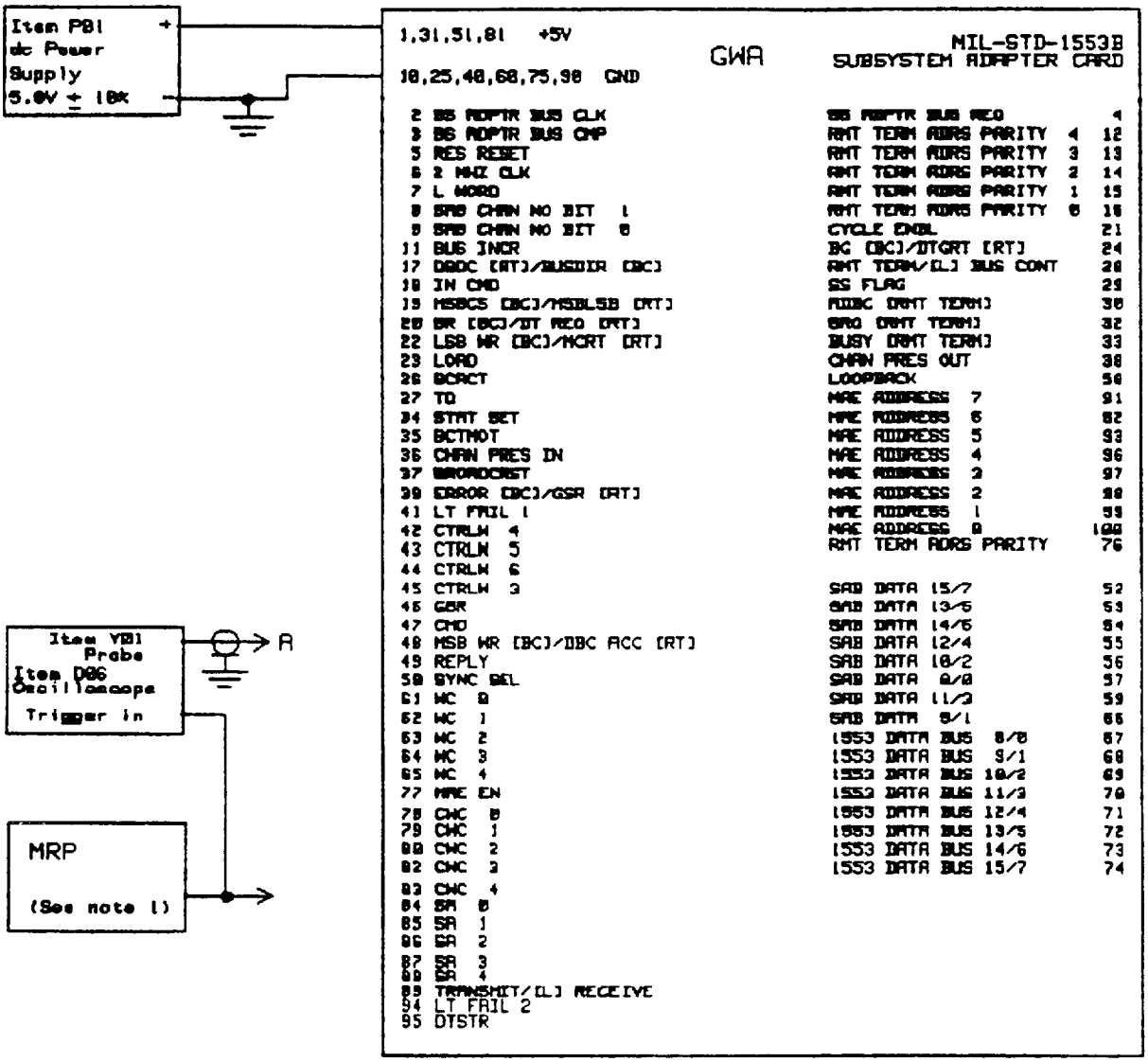


FIGURE 14. Power supply transient amplitude waveform for table XX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XX.

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